Where Should the Code Go? Locating Foundation-Model Errors in Bitemporal Inventory Simulation

Durga Sandeep Saluru
Distyl AI
durga@distyl.ai

Lanchen Song
Distyl AI
lanchen.song@distyl.ai

Aditya Kumaran
Distyl AI
aditya.kumaran@distyl.ai

Abstract

Operational reports carry two clocks: when an event happened and when its report arrived; later reports can revise earlier ones. A foundation-model component that maintains the state supported by the reports visible at a cutoff, then rolls a supplied scenario forward, can be helped by exact code before it, after it or inside the call, or asked to reason harder. We ask which remedy removes which error, for which models, on an exactly solvable lost-sales inventory environment: 36,672 independently rescored requests over twelve configurations of closed and open-weight models, with a forced-answer-call interface as a control and protocols frozen before the calls. Weak models fail in separable ways. Canonical input removes report-selection error, exact continuation removes rollout error, neither repairs the other, and which error dominates differs by model; strong models are at ceiling on raw reports. Asking for more reasoning is no substitute and can depend on the interface: under a forced answer call high effort collapses GPT-5.4 mini and DeepSeek V4 Pro, which gain or hold under automatic choice; on its provider the forced call disables reasoning for DeepSeek V4 Flash; and at high effort gpt-oss exhausts its output budget. The contribution is a controlled component diagnostic, released with every request, and evidence that for some model and provider pairs the tool-calling interface confounds reasoning evaluation.

1 Motivation and related work

The state supported by the reports visible at a knowledge cutoff K is therefore not the physical state, and reconstructing it requires an arrival-time cutoff, revision resolution, and replay in event-time order. Lost-sales dynamics make order consequential: unmet demand is lost, so swapping a receipt and a demand changes the trajectory at equal totals. We study a component that maintains this temporal state from an event stream and then applies an explicitly supplied future schedule, which separates state error from forecasting uncertainty and makes the reporting cutoff a leakage control rather than a nuisance.

Ordinary code solves this task exactly, yet models hold operational state in agentic pipelines: inventory agents reason over order and arrival histories in context [1], and their stated beliefs and actions can diverge [5]. We ask where the designer of such a pipeline should put exact code, and which placement removes which error for which models, when every intermediate quantity has an executable ground truth; the seeded environment and every request and response are released.[1]

Related work. Temporal QA benchmarks test when facts hold (valid time) [4, 12–14, 18, 19]; BiTempQA annotates event and record time for QA [10]. None asks a model to reconstruct a numeric

[1] https://github.com/durga-sandeep/where-should-the-code-go

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Foundation Models for Temporal Systems.

state *as of* a record time under revisions, and the inventory benchmarks that do use lost-sales dynamics evaluate ordering policy, not ledger replay [1, 5]. Program-aided prompting puts exact execution after the model [3, 8] and code-interpreter agents put it inside the call, trading arithmetic error for other errors [7]; the three-way comparison with per-stage error attribution is, to our knowledge, new. On interfaces, format constraints are known to tax reasoning, most for weaker models [2, 6, 16], and more test-time reasoning can lower accuracy [9]; we find that for some model and provider pairs (clearly GPT-5.4 mini and DeepSeek V4 Pro, slightly GPT-5.4) the format tax grows with requested effort under a forced answer call, while for four others it does not. State tracking degrades with depth across scales [11, 15]; our model ladder is only the substrate here.

2 Environment and study design

Semantics. A report has a logical event identifier, revision, event time, received time, kind and quantity. For target time T and cutoff K: keep reports received by K, retain the highest visible revision per event, keep events with event time at most T, and replay in (event time, identifier) order from the initial stock. A receipt of q adds q; a demand of q sets $x' = \max(0, x - q)$ and loses $\max(0, q - x)$. A supplied schedule through horizon H is then applied with the same rule. The full output is five integers: stock at T, historical loss, stock at H, future loss, and the first future time with positive loss (-1 if none; formal rule and worked example in Appendix A).

Where the code sits. Three full-output arms move executable work before the model. **Raw** supplies every report version in received-time order, including reports after K that must be ignored; **canonical** supplies the correctly selected, version-resolved events in event-time order; **snapshot** additionally supplies the exactly replayed historical state, so only the rollout remains. Two **state-only** arms move executable work after the model: with raw or canonical input and no schedule, the model returns only stock at T and historical loss, and code applies the schedule to the emitted stock, giving a five-field *hybrid* answer. Two **code-inside** arms give the OpenAI configurations a hosted code tool, with raw or canonical input and the full output; the model chooses whether to run code. A post-hoc **continuation** applies the same code to the stock emitted in the full-output arms; by conservation of units it is exact exactly when that stock is, so hybrid future accuracy equals reconstruction accuracy (an identity, verified on every answer).

Configurations, cohorts and statistics. Twelve configurations: GPT-5.4 mini and GPT-5.4 (OpenAI), Claude Haiku 4.5 and Sonnet 4.6 each without and with extended thinking (Anthropic), and six open-weight models through OpenRouter, each pinned to one upstream provider: gpt-oss-20b and gpt-oss-120b, Qwen3.8-27B, DeepSeek V4 Flash and V4 Pro, and GLM-5.3 Flash (3.6B to 49B active parameters). The main table uses automatic tool choice at the lowest requested effort for every model; the OpenAI and open-weight models are also run under a forced answer call (the answer function is the only permitted action) at requested effort low and high, giving a 2×2 effort by interface cross per model. Requests use a strict integer tool schema, 16,384 output tokens and each provider's default sampling (we set no temperature or top-p), with no retries of any model output; missing, malformed, refused or truncated outputs fail. The synthetic cohort has 96 seeded heldout histories in a 24-cell factorial over history length, reporting delay, corrections and schedule type (Appendix A). The retail-profile cohort maps 48 public FreshRetailNet [17] store windows to demand quantities; stock, receipts, delays and revisions are simulated. Each configuration/arm/history cell has two requests. Endpoints are all-five exactness and *future-joint* exactness (all three scenario outputs). Paired differences average repetitions within history and use 10,000 history-bootstrap percentile intervals, reported as descriptive 95% intervals without multiplicity adjustment. They do not capture variation between collection periods: re-running one identical cell several days later moved it by 8.9 points, so we treat smaller differences as unresolved. Every study was fixed in a written protocol before its calls; independent re-parsing and re-scoring of every response found zero mismatches (Appendices E and F; controls in Appendix C).

3 Results

Code before the model removes selection error and most replay error, not rollout error. Table 1 reports every configuration under one interface; 6 of 12 start below 90% on raw reports and the rest are near ceiling there, so placement is a question about weak components. For the weak ones,

	Synthetic				Retail profiles		
Model (automatic tool choice)	raw	canonical	hybrid	snapshot	raw	canonical	snapshot
GPT-5.4 mini	56.8	86.5	94.8	94.8	14.6	82.3	89.6
Haiku 4.5	52.6	85.9	92.7	92.7	6.2	89.6	92.7
Haiku 4.5 + thinking	88.5	96.9	100.0	99.0	40.6	91.7	96.9
gpt-oss-20b†	24.0	51.0	71.4	82.8	0.0	17.7	77.1
gpt-oss-120b†	49.5	73.4	76.0	94.8	5.2	72.9	86.5
Qwen3.8-27B†	95.3	99.5	100.0	100.0	96.9	99.0	100.0
DeepSeek V4 Flash†	98.4	99.0	99.0	99.0	94.8	96.9	99.0
GLM-5.3 Flash†	55.7	64.1	87.0	58.9	26.0	58.3	38.5
DeepSeek V4 Pro†	99.0	99.0	99.0	99.0	95.8	100.0	95.8
GPT-5.4	98.4	98.4	83.9	91.1	97.9	100.0	89.6
Sonnet 4.6	99.0	99.5	100.0	100.0	95.8	99.0	100.0
Sonnet 4.6 + thinking	97.9	100.0	100.0	100.0	97.9	100.0	100.0

Table 1: Future-joint accuracy (%, all three scenario outputs exact) by input and code placement, every configuration under automatic tool choice at its lowest requested effort (OpenAI and open-weight models: requested effort low; Anthropic models: no or 4,096-token thinking); 192 synthetic and 96 profile calls per cell; invalid and missing answers fail. *raw*: model selects, replays and rolls out; *canonical*: code selects and orders eligible events; *hybrid*: model returns the two historical fields from canonical input and code applies the schedule (equals reconstruction accuracy by conservation); *snapshot*: exact state supplied, model rolls out. †Open-weight models through OpenRouter, each pinned to one upstream provider (Appendix F). Forced-call cells and all-five accuracy: Appendix B.

canonical input raises future-joint accuracy by mini +29.7 [+20.8, +38.5]; Haiku +33.3 [+24.0, +43.2]; Haiku+thinking +8.3 [+3.1, +14.1]; gpt-oss-20b +27.1 [+19.8, +34.4]; gpt-oss-120b +24.0 [+16.1, +31.8]. Exact state does not make them exact: with only the 12-event rollout left, every weak model stays below 100% on snapshot input (Table 1). The decomposition also locates different errors in different models. For GLM canonical input adds +8.3 [+0.0, +16.7] and exact state adds nothing over it (-5.2 [-14.6, +4.7]), yet code after the model on the same canonical answers adds +22.9 [+15.1, +30.7]: its error is almost entirely rollout. For gpt-oss-20b both stages fail (+27.1 [+19.8, +34.4] from canonical input, +31.8 [+21.4, +42.2] more from exact state).

Code after the model removes rollout error but cannot repair state. This holds by construction (Section 2); what is empirical is how much each model gains. Run prospectively, the state-only hybrid's gain over the model's own rollout on canonical input is mini +8.3 [+3.6, +13.5]; Haiku +6.8 [+0.0, +13.5]; gpt-oss-20b +20.3 [+12.0, +28.6]; gpt-oss-120b +2.6 [-4.2, +9.4]; GLM +22.9 [+15.1, +30.7]. On raw input the hybrid never exceeds 78% for a weak model: code after the model removes rollout error wherever the reconstruction was right and cannot repair a stock that is wrong because the wrong reports were selected.

Code inside the call. With a hosted code-execution tool available (OpenAI configurations only), mini's raw-input future-joint accuracy rises by +54.2 [+45.3, +62.5] points on synthetic histories and +77.1 [+68.8, +84.4] on profiles, and GPT-5.4 reaches 100% on raw input in both cohorts; mini ran code in 469 of 576 calls and was exact in 453 of them, versus 93 of the 107 calls it answered without code. Against the interface-matched automatic-choice baseline the tool's own contribution on synthetic histories is +34.9 [+26.0, +43.8] points on raw and +11.5 [+6.2, +17.7] on canonical input. Letting the model delegate selection and rollout to code it writes is the most effective single placement at fixed input, though not error-free (Appendix E).

Reasoning harder fails in three interface-dependent ways. Table 2 crosses requested effort with the tool-calling interface. For DeepSeek V4 Flash on its provider the forced call effectively disabled reasoning: forced canonical calls averaged 32 reasoning tokens against 1695 under automatic choice, and accuracy is 2.1 against 99.0%; effort changes nothing under either interface. For GPT-5.4 mini and for DeepSeek V4 Pro on its provider the forced call interacts with effort: mini's high-minus-low effect on canonical input is -41.1 [-51.0, -31.2] under the forced call and +6.8 [+0.5, +13.5] under automatic choice, and DeepSeek V4 Pro falls from 98.4 to 82.3% under the forced call at high effort (32 of 192 calls with no answer) while staying at 97.4% under automatic choice. GPT-5.4 shows a small interaction of the same sign (+5.2 [+1.0, +9.4]). For gpt-oss-20b, gpt-oss-120b, Qwen3.8-27B and GLM-5.3 Flash the interaction interval covers zero: effort acts the same way under both interfaces.

	Forced answer call			Automatic tool choice			
Model	low	high	high−low	low	high	high−low	Interaction
GPT-5.4 mini	76.0	34.9	-41.1	86.5	93.2	+6.8	+47.9 [+35.9, +59.9]
GPT-5.4	98.4	94.8	-3.6	98.4	100.0	+1.6	+5.2 [+1.0, +9.4]
gpt-oss-20b	52.1	66.1	+14.1	51.0	59.4	+8.3	-5.7 [-15.6, +4.2]
gpt-oss-120b	69.3	66.1	-3.1	73.4	68.8	-4.7	-1.6 [-10.4, +6.8]
Qwen3.8-27B	91.7	91.7	+0.0	99.5	92.7	-6.8	-6.8 [-14.1, +0.5]
DeepSeek V4 Flash	2.1	1.0	-1.0	99.0	97.4	-1.6	-0.5 [-4.7, +3.1]
GLM-5.3 Flash	62.5	82.3	+19.8	64.1	84.4	+20.3	+0.5 [-9.9, +11.5]
DeepSeek V4 Pro	98.4	82.3	-16.1	99.0	97.4	-1.6	+14.6 [+8.3, +20.8]

Table 2: Requested reasoning effort $\times$ tool-choice interface, synthetic cohort, canonical input, future-joint accuracy (%; 192 calls per cell). high−low: paired effort effect within an interface. Interaction: automatic-choice effort effect minus forced-call effort effect, paired by history, with a descriptive 95% history-bootstrap interval (10,000 draws); a positive interaction means the forced call turns extra effort into a liability. Requested effort is a family-specific provider setting, not a compute match (observed reasoning tokens: Appendix F).

A third failure of *reason harder* is budget exhaustion: at high effort gpt-oss-120b spent the whole 16,384-token output limit on reasoning in 65 of 192 forced canonical calls and gpt-oss-20b in 97 of 192 automatic raw calls, each a missing answer. gpt-oss-20b also refused the fictional task (88 of 192 low-effort raw calls had no answer, mostly refusals), so its cells measure willingness as much as ability. Reasoning tokens and missing answers for every cell: Appendix F.

State and decision errors differ (post hoc). For mini, 166 of 208 wrong starting stocks still give the right stockout decision (Appendix D).

4 Limitations and implications

The task is solved exactly by code; the results say where to place code around a model that already holds operational state, not whether a model should replace code. Synthetic histories, artificial corrections, two repetitions, one retail source and 96 histories limit coverage; differences under nine points, including GLM's canonical gain and Haiku's hybrid gain, are not resolved. Snapshot input is not a pure rollout probe for every model: GLM, and DeepSeek V4 Pro on retail profiles, score lower with exact state than with canonical input, so the state's presentation may itself change behaviour. Each open-weight model is served by one upstream provider whose quantization and effort mapping we do not control, so its interface effects are claims about the model and provider pair, not the model; missing answers, refusals and truncations count as failures, so some cells (gpt-oss-20b, high-effort gpt-oss-120b) measure completion as much as accuracy. GPT-5.4 under automatic choice scored lower on the snapshot and hybrid arms than under the forced call collected a day earlier while its raw and canonical cells did not move; a same-period forced-call replication of those two arms (Appendix F) moved them by -1.0 and +0.5 points relative to the original cells, so the automatic-choice difference against the contemporaneous baseline is -6.8 and -15.1 points.What the diagnostic supports is narrower than its title: for weak components, reconstruction and rollout fail separately and code can be placed to remove each; for strong components the placement is moot; and an evaluation that raises reasoning effort through a forced tool call, or compares families under different tool-calling interfaces, can measure the interface as much as the model, as it clearly did here for three of the eight pairs crossed.

Pipelines with a time-series foundation model (TSFM). No TSFM is evaluated; forecasting was removed to measure state error exactly. Upstream, when an LLM reconstructs the history a TSFM consumes from late and revised reports, selection and replay errors become forecast inputs no forecaster can repair; normalizing the reports in code first removes them. Downstream, when a TSFM forecast supplies the schedule, the rollout our snapshot and hybrid arms isolate belongs in code, so rollout error is not mistaken for forecast error. The cutoff K is also the leakage control a forecasting backtest needs: scoring on revisions not visible at K overstates accuracy.

AI assistance. AI assistants were used extensively in this work; the authors are responsible for all claims.

References

[1] Jackie Baek, Yaopeng Fu, Will Ma, and Tianyi Peng. AI agents for inventory control: Human-LLM-OR complementarity. arXiv:2602.12631, 2026. URL https://arxiv.org/abs/2602.12631.

[2] Debangshu Banerjee, Tarun Suresh, Shubham Ugare, Sasa Misailovic, and Gagandeep Singh. CRANE: Reasoning with constrained LLM generation. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2502.09061.

[3] Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research*, 2023. URL https://openreview.net/forum?id=YfZ4ZPt8zd.

[4] Zheng Chu, Jingchang Chen, Qianglong Chen, Weijiang Yu, Haotian Wang, Ming Liu, and Bing Qin. TimeBench: A comprehensive evaluation of temporal reasoning abilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1204–1228, 2024. URL https://aclanthology.org/2024.acl-long.66/.

[5] Sagar Deb and Ashwanth Krishnan. STOCKTAKE: Measuring the gap between perception and action in LLM agents with a fair oracle. arXiv:2607.13618, 2026. URL https://arxiv.org/abs/2607.13618.

[6] Hengxin Fan. Capacity, not format: Rethinking structured reasoning failures. arXiv:2606.09410, 2026. URL https://arxiv.org/abs/2606.09410.

[7] Farima Fatahi Bayat, Pouya Pezeshkpour, and Estevam Hruschka. From proof to program: Characterizing tool-induced reasoning hallucinations in large language models. arXiv:2511.10899, 2025. URL https://arxiv.org/abs/2511.10899.

[8] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. URL https://proceedings.mlr.press/v202/gao23f.html.

[9] Aryo Pradipta Gema, Alexander Hong, Milena Mojsilović, et al. Inverse scaling in test-time compute, 2025. URL https://arxiv.org/abs/2507.14417. arXiv:2507.14417.

[10] heihei. BiTempQA: A diagnostic benchmark for bitemporal reasoning in LLM agent memory systems. Public Hugging Face dataset card, 2026. URL https://huggingface.co/datasets/heihei/BiTempQA. Accessed 7 September 2026; peer-reviewed publication status unverified.

[11] Dengzhe Hou, Lingyu Jiang, Deng Li, Zirui Li, Fangzhou Lin, and Kazunori D. Yamada. WMF-AM: Probing LLM working memory via depth-parameterized cumulative state tracking. arXiv:2603.27343, 2026. URL https://arxiv.org/abs/2603.27343.

[12] Soyeon Kim, Jindong Wang, Xing Xie, and Steven Euijong Whang. Harnessing temporal databases for systematic evaluation of factual time-sensitive question-answering in large language models. In *Proceedings of the 14th International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2508.02045.

[13] Sigang Luo, Yinan Liu, Dongying Lin, Yingying Zhai, Bin Wang, Xiaochun Yang, and Junpeng Liu. ETRQA: A comprehensive benchmark for evaluating event temporal reasoning abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23321–23339, 2025. URL https://aclanthology.org/2025.findings-acl.1198/.

[14] Nishanth Sridhar Nakshatri, Shamik Roy, Manoj Ghuhan Arivazhagan, Hanah Zhou, Vinayshekhar Bannihatti Kumar, and Rashmi Gangadharaiah. When facts change: Probing LLMs on evolving knowledge with evolveQA. arXiv:2510.19172, 2025. URL https://arxiv.org/abs/2510.19172.

[15] Akshit Sinha, Arvindh Arun, Shashwat Goel, Steffen Staab, and Jonas Geiping. The illusion of diminishing returns: Measuring long horizon execution in LLMs. In *Proceedings of the 14th International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2509.09677.

[16] Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1218–1236, 2024. URL https://arxiv.org/abs/2408.02442.

[17] Yangyang Wang, Jiawei Gu, Li Long, Xin Li, Li Shen, Zhouyu Fu, Xiangjun Zhou, and Xu Jiang. FreshRetailNet-50K: A stockout-annotated censored demand dataset for latent demand recovery and forecasting in fresh retail. arXiv:2505.16319, 2025. URL https://arxiv.org/abs/2505.16319.

[18] Shaohang Wei, Wei Li, Feifan Song, Wen Luo, Tianyi Zhuang, Haochen Tan, Zhijiang Guo, and Houfeng Wang. TimE: A multi-level benchmark for temporal reasoning of LLMs in real-world scenarios. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/84f1e188c2be52f89f6e206bc37d092d-Abstract-Datasets_and_Benchmarks_Track.html.

[19] Muyan Weng, Defu Cao, Wei Yang, Yashaswi Sharma, and Yan Liu. TemporalBench: A benchmark for evaluating LLM-based agents on contextual and event-informed time series tasks. arXiv:2602.13272, 2026. URL https://arxiv.org/abs/2602.13272.

A Executable semantics, provider settings and sampling

The task reconstructs the state supported by a versioned report history, rather than recovering unobserved physical inventory. For each report r, let $e(r)$ be its logical event identifier, $v(r)$ its revision, $t(r)$ its event time and $a(r)$ its received time. The selected version is

$$r_e^*(K) = \arg \max_{r:\, e(r)=e,\, a(r)\le K} v(r).$$

Identifiers with no visible report are absent. Only selected versions with $t(r_e^*) \le T$ are replayed. Thus a newer visible revision outside the target horizon does not cause an older revision to be substituted. This edge case is covered by unit tests, although generated experimental historical event times all precede T. The generator uses distinct revisions, so revision ties do not arise. Replay sorts by event time and then event identifier.

Illustrative two-clock example. Start with two units of stock. A receipt of five units occurs at event time 1, but its report arrives at time 4. A demand of six units occurs and is reported at time 2. Reconstruct through $T = 3$. With $K = 3$, only the demand is visible: the reconstructed stock is zero and historical loss is four. With $K = 4$, replay the receipt before the demand: reconstructed stock is one and historical loss is zero. A supplied future demand of three units at time 5 then loses three units in the first view and two in the second. Both views finish at zero stock and have first future stockout time 5. The later knowledge view revises an earlier reconstruction; it is not a forecast that had access to later reports.

Resource and output contract. OpenAI requests use dated model identifiers `gpt-5.4-mini-2026-03-17` and `gpt-5.4-2026-03-05`, requested reasoning effort low (medium and high for the effort sweep), and forced `submit_answer` tool choice. Both Anthropic configurations use `claude-sonnet-4-6`, automatic tool choice, and disabled parallel tool use; one additionally requests manual extended thinking with a 4096-token budget. The budget is a requested thinking target, not a strict observed token ceiling. Full arms use a strict five-integer schema; state-only arms use a strict two-integer schema and instructions that replace the future-scenario paragraph with an explicit statement that no future is to be simulated. All requests allow 16384 total output tokens. Automatic tool choice may yield reasoning text without an answer; absent or malformed answers remain failures. Requests have a 420-second transport timeout and are not retried or repaired. Exact payloads, returned token usage and provider responses are included in the supplement. These interfaces and compute allowances are disclosed rather than treated as a controlled cross-provider comparison.

Sampling and uncertainty. The primary cohort contains 2304 requests, the later-knowledge control 576, the display-order control 192, and the profile cohort 1152, for 4224 in total. The effort sweeps add 3,648 requests, the state-only arms 2,880, the code-inside arms 1,152, and the replication 576, all on the same histories. The controls reuse the first 24 primary histories. Each of the 96 primary histories and 48 profile windows has two repetitions per configuration and arm. Repetition averages are paired within the same history/window before bootstrap resampling. Intervals characterize variation within this constructed sample; they are not population guarantees or corrections for multiple comparisons. Source-store separation does not establish independence of retail behavior across stores. Extension studies add configurations and arms on the archived histories, not independent histories. Oracle task coverage counts each base once, independently of model calls; these are scenario properties, not model success rates. The synthetic cohort has 52/96 bases with positive historical loss, 61/96 with positive future loss, and 56/96 with positive terminal stock. These categories can overlap. The external-profile cohort has 28/48 bases with positive historical loss, 25/48 with positive future loss, and 37/48 with positive terminal stock. These categories can overlap.

Overlap and contamination scope. A post-hoc exact-content audit finds no duplicates within, or overlap between, 24 regenerated v2 development histories and the 96 heldout histories, or between 16 development and 48 heldout profile tasks. Profile source stores are disjoint; exact ordered quantized sales sequences also have no overlap. Checks use frozen generators and are archived with their scope. Intentional control/repetition reuse is excluded. These checks do not test near-duplicates or provider training data; the public sales profiles may have appeared in training. No cases were removed or rescored by this audit.

Configuration	Raw	Canonical	Snapshot
GPT-5.4 mini (low)	32.8 / 37.5	61.5 / 76.0	89.6 / 89.6
GPT-5.4 (low)	83.9 / 88.0	94.8 / 98.4	99.0 / 99.0
Sonnet 4.6	97.9 / 99.0	99.0 / 99.5	100.0 / 100.0
Sonnet + thinking	96.9 / 97.9	100.0 / 100.0	100.0 / 100.0

Table 3: Held-out synthetic replay: each cell is all-five / future-joint exact accuracy (%), 192 calls per configuration/protocol. The first endpoint requires all five outputs; the second requires all three future-scenario outputs. Invalid contracts fail. Thinking denotes a separate configuration.

Configuration	Raw	Canonical	Snapshot
GPT-5.4 mini (low)	4.2 / 12.5	40.6 / 46.9	89.6 / 89.6
GPT-5.4 (low)	88.5 / 90.6	91.7 / 91.7	99.0 / 99.0
Sonnet 4.6	91.7 / 95.8	97.9 / 99.0	100.0 / 100.0
Sonnet + thinking	96.9 / 97.9	100.0 / 100.0	100.0 / 100.0

Table 4: External-profile simulation: each cell is all-five / future-joint exact accuracy (%), 96 calls per configuration/protocol. The first endpoint requires all five outputs; the second requires all three future-scenario outputs. Invalid contracts fail. Thinking denotes a separate configuration.

Retail-profile time axis. Source hours determine ordered two-hour sales bins. The adapter assigns synthetic event times to alternating historical receipts/demands and interleaved future receipts/demands; these event-time units are not literal calendar hours. The profile check preserves sequential quantity patterns rather than validating real operational reporting timestamps.

B All-five accuracy tables

Tables 3 and 4 report both the all-five and the future-joint endpoints for the original grid; Table 5 stratifies mini's synthetic results by design factor; Table 6 lists every cell including the extension studies. Figures 2 and 3 plot the original grid's historical and future error rates and observed token cost by representation.

Stratum	Stock exact		Future joint	
	raw	canonical	raw	canonical
immediate reporting	63/96	86/96	46/96	71/96
delayed reporting	34/96	83/96	26/96	75/96
no corrections	62/96	91/96	46/96	76/96
corrections present	35/96	78/96	26/96	70/96
12 events	50/64	61/64	43/64	59/64
36 events	26/64	59/64	18/64	51/64
72 events	21/64	49/64	11/64	36/64

Table 5: GPT-5.4 mini (low) on the synthetic factorial, by design factor (96 calls per stratum). Strata are different seeded histories, not matched interventions; counts are descriptive.

Table 6: All cells of the original grid and the extension studies (%). State-only arms report the hybrid answer (model historical fields plus executable continuation); their future-joint accuracy equals their stock accuracy by conservation. Tool arms had a code-execution tool available. Invalid answers (missing, malformed or negative) fail every endpoint and remain in denominators.

Cohort	Configuration	Arm	N	Invalid	Missing	All five	Stock	Hist. joint	Future joint
Synthetic	GPT-5.4 mini (low)	raw	192	0	0	32.8	50.5	43.2	37.5
Synthetic	GPT-5.4 mini (low)	canonical	192	0	0	61.5	88.0	69.3	76.0
Synthetic	GPT-5.4 mini (low)	snapshot	192	0	0	89.6	100.0	100.0	89.6
Synthetic	GPT-5.4 mini (low)	state-raw	192	0	0	43.2	55.2	43.2	55.2
Synthetic	GPT-5.4 mini (low)	state-canonical	192	0	0	73.4	93.8	73.4	93.8
Synthetic	GPT-5.4 mini (low)	tool-raw	192	0	0	90.1	94.8	91.1	91.7
Synthetic	GPT-5.4 mini (low)	tool-canonical	192	0	0	94.8	99.5	95.8	97.9
Synthetic	GPT-5.4 mini (medium)	raw	192	2	0	23.4	45.8	35.4	26.6
Synthetic	GPT-5.4 mini (medium)	canonical	192	1	0	43.2	66.1	58.3	46.9

continued on next page

Table 6 (continued)

Cohort	Configuration	Arm	N	Invalid	Missing	All five	Stock	Hist. joint	Future joint
Synthetic	GPT-5.4 mini (medium)	snapshot	192	0	0	97.9	100.0	100.0	97.9
Synthetic	GPT-5.4 mini (high)	raw	192	0	0	22.4	41.7	34.4	25.0
Synthetic	GPT-5.4 mini (high)	canonical	192	1	0	33.9	56.2	52.1	34.9
Synthetic	GPT-5.4 mini (high)	snapshot	192	0	0	97.4	100.0	100.0	97.4
Synthetic	GPT-5.4 mini (high)	state-raw	192	0	0	45.3	54.2	45.3	54.2
Synthetic	GPT-5.4 mini (high)	state-canonical	192	1	0	64.6	71.4	64.6	71.4
Synthetic	GPT-5.4 mini (low, auto tool choice)	raw	192	1	1	46.4	58.9	47.9	56.8
Synthetic	GPT-5.4 mini (low, auto tool choice)	canonical	192	0	0	72.4	91.7	76.6	86.5
Synthetic	GPT-5.4 mini (low, auto tool choice)	snapshot	192	0	0	94.8	100.0	100.0	94.8
Synthetic	GPT-5.4 mini (low, auto tool choice)	state-raw	192	0	0	53.6	69.3	53.6	69.3
Synthetic	GPT-5.4 mini (low, auto tool choice)	state-canonical	192	0	0	80.2	94.8	80.2	94.8
Synthetic	GPT-5.4 mini (high, auto tool choice)	raw	192	1	1	72.4	82.3	75.5	79.2
Synthetic	GPT-5.4 mini (high, auto tool choice)	canonical	192	0	0	89.1	96.9	92.7	93.2
Synthetic	GPT-5.4 mini (high, auto tool choice)	snapshot	192	0	0	98.4	100.0	100.0	98.4
Synthetic	GPT-5.4 (low)	raw	192	0	0	83.9	91.1	86.5	88.0
Synthetic	GPT-5.4 (low)	canonical	192	0	0	94.8	99.5	95.8	98.4
Synthetic	GPT-5.4 (low)	snapshot	192	0	0	99.0	100.0	100.0	99.0
Synthetic	GPT-5.4 (low)	state-raw	192	0	0	88.5	91.7	88.5	91.7
Synthetic	GPT-5.4 (low)	state-canonical	192	0	0	96.9	98.4	96.9	98.4
Synthetic	GPT-5.4 (low)	tool-raw	192	0	0	100.0	100.0	100.0	100.0
Synthetic	GPT-5.4 (low)	tool-canonical	192	0	0	96.9	98.4	97.9	97.4
Synthetic	GPT-5.4 (medium)	raw	192	0	0	95.3	98.4	98.4	95.3
Synthetic	GPT-5.4 (medium)	canonical	192	0	0	93.8	97.9	95.3	95.8
Synthetic	GPT-5.4 (high)	raw	192	0	0	94.3	96.9	96.4	94.3
Synthetic	GPT-5.4 (high)	canonical	192	0	0	94.3	98.4	97.9	94.8
Synthetic	GPT-5.4 (low, auto tool choice)	raw	192	0	0	96.4	98.4	96.4	98.4
Synthetic	GPT-5.4 (low, auto tool choice)	canonical	192	0	0	95.8	98.4	95.8	98.4
Synthetic	GPT-5.4 (low, auto tool choice)	snapshot	192	0	0	91.1	100.0	100.0	91.1
Synthetic	GPT-5.4 (low, auto tool choice)	state-raw	192	0	0	96.4	96.9	96.4	96.9
Synthetic	GPT-5.4 (low, auto tool choice)	state-canonical	192	0	0	81.2	83.9	81.2	83.9
Synthetic	GPT-5.4 (high, auto tool choice)	raw	192	0	0	99.0	99.5	99.5	99.0
Synthetic	GPT-5.4 (high, auto tool choice)	canonical	192	0	0	99.5	100.0	99.5	100.0
Synthetic	Haiku 4.5	raw	192	0	0	43.8	59.4	47.9	52.6
Synthetic	Haiku 4.5	canonical	192	0	0	79.2	96.9	87.0	85.9
Synthetic	Haiku 4.5	snapshot	192	0	0	92.7	100.0	100.0	92.7
Synthetic	Haiku 4.5	state-raw	192	0	0	44.3	61.5	44.3	61.5
Synthetic	Haiku 4.5	state-canonical	192	0	0	86.5	92.7	86.5	92.7
Synthetic	Haiku 4.5 + thinking	raw	192	10	10	82.3	90.1	83.3	88.5
Synthetic	Haiku 4.5 + thinking	canonical	192	0	0	94.8	100.0	96.4	96.9
Synthetic	Haiku 4.5 + thinking	snapshot	192	0	0	99.0	100.0	100.0	99.0
Synthetic	Haiku 4.5 + thinking	state-raw	192	11	11	84.9	88.0	84.9	88.0
Synthetic	Haiku 4.5 + thinking	state-canonical	192	0	0	99.0	100.0	99.0	100.0
Synthetic	Sonnet 4.6	raw	192	0	0	97.9	99.5	98.4	99.0
Synthetic	Sonnet 4.6	canonical	192	0	0	99.0	100.0	99.5	99.5
Synthetic	Sonnet 4.6	snapshot	192	0	0	100.0	100.0	100.0	100.0
Synthetic	Sonnet 4.6	state-raw	192	0	0	99.5	100.0	99.5	100.0
Synthetic	Sonnet 4.6	state-canonical	192	0	0	99.0	100.0	99.0	100.0
Synthetic	Sonnet + thinking	raw	192	0	0	96.9	99.0	97.9	97.9
Synthetic	Sonnet + thinking	canonical	192	0	0	100.0	100.0	100.0	100.0
Synthetic	Sonnet + thinking	snapshot	192	0	0	100.0	100.0	100.0	100.0
Synthetic	Sonnet + thinking	state-raw	192	0	0	99.0	100.0	99.0	100.0
Synthetic	Sonnet + thinking	state-canonical	192	0	0	99.5	100.0	99.5	100.0
Synthetic	gpt-oss-20b (low, forced call)	raw	192	85	85	20.3	31.2	26.6	23.4
Synthetic	gpt-oss-20b (low, forced call)	canonical	192	59	59	50.5	58.3	54.7	52.1
Synthetic	gpt-oss-20b (low, auto tool choice)	raw	192	88	88	20.8	31.8	25.5	24.0
Synthetic	gpt-oss-20b (low, auto tool choice)	canonical	192	51	51	46.4	62.5	52.1	51.0
Synthetic	gpt-oss-20b (low, auto tool choice)	snapshot	192	0	0	82.8	100.0	100.0	82.8
Synthetic	gpt-oss-20b (low, auto tool choice)	state-raw	192	53	53	29.7	37.5	29.7	37.5
Synthetic	gpt-oss-20b (low, auto tool choice)	state-canonical	192	32	32	57.8	71.4	57.8	71.4
Synthetic	gpt-oss-20b (high, forced call)	raw	192	106	106	42.2	42.2	42.2	42.2
Synthetic	gpt-oss-20b (high, forced call)	canonical	192	65	65	66.1	66.1	66.1	66.1
Synthetic	gpt-oss-20b (high, auto tool choice)	raw	192	112	112	39.1	40.6	40.6	39.1
Synthetic	gpt-oss-20b (high, auto tool choice)	canonical	192	75	75	59.4	60.4	60.4	59.4
Synthetic	gpt-oss-120b (low, forced call)	raw	192	0	0	42.7	53.6	46.4	45.8
Synthetic	gpt-oss-120b (low, forced call)	canonical	192	0	0	62.0	79.7	68.8	69.3
Synthetic	gpt-oss-120b (low, auto tool choice)	raw	192	41	41	46.9	54.7	50.5	49.5
Synthetic	gpt-oss-120b (low, auto tool choice)	canonical	192	12	12	67.7	78.1	69.8	73.4
Synthetic	gpt-oss-120b (low, auto tool choice)	snapshot	192	1	1	94.8	99.0	99.0	94.8
Synthetic	gpt-oss-120b (low, auto tool choice)	state-raw	192	16	16	48.4	57.8	48.4	57.8
Synthetic	gpt-oss-120b (low, auto tool choice)	state-canonical	192	4	4	68.2	76.0	68.2	76.0
Synthetic	gpt-oss-120b (high, forced call)	raw	192	89	89	53.6	53.6	53.6	53.6
Synthetic	gpt-oss-120b (high, forced call)	canonical	192	65	65	66.1	66.1	66.1	66.1
Synthetic	gpt-oss-120b (high, auto tool choice)	raw	192	98	98	49.0	49.0	49.0	49.0
Synthetic	gpt-oss-120b (high, auto tool choice)	canonical	192	60	60	68.8	68.8	68.8	68.8
Synthetic	Qwen3.8-27B (low, forced call)	raw	192	41	40	75.5	77.1	76.0	76.6
Synthetic	Qwen3.8-27B (low, forced call)	canonical	192	16	16	91.7	91.7	91.7	91.7
Synthetic	Qwen3.8-27B (low, auto tool choice)	raw	192	3	3	92.2	95.3	92.2	95.3

continued on next page

Table 6 (continued)

Cohort	Configuration	Arm	N	Invalid	Missing	All five	Stock	Hist. joint	Future joint
Synthetic	Qwen3.8-27B (low, auto tool choice)	canonical	192	0	0	98.4	99.5	98.4	99.5
Synthetic	Qwen3.8-27B (low, auto tool choice)	snapshot	192	0	0	100.0	100.0	100.0	100.0
Synthetic	Qwen3.8-27B (low, auto tool choice)	state-raw	192	3	3	95.8	96.9	95.8	96.9
Synthetic	Qwen3.8-27B (low, auto tool choice)	state-canonical	192	0	0	99.5	100.0	99.5	100.0
Synthetic	Qwen3.8-27B (high, forced call)	raw	192	28	28	77.1	81.8	78.1	80.2
Synthetic	Qwen3.8-27B (high, forced call)	canonical	192	10	10	90.6	94.3	93.2	91.7
Synthetic	Qwen3.8-27B (high, auto tool choice)	raw	192	19	19	85.9	88.5	85.9	88.5
Synthetic	Qwen3.8-27B (high, auto tool choice)	canonical	192	8	8	91.1	95.8	94.3	92.7
Synthetic	DeepSeek V4 Flash (low, forced call)	raw	192	8	4	2.1	6.2	2.1	2.1
Synthetic	DeepSeek V4 Flash (low, forced call)	canonical	192	10	6	2.1	7.8	2.6	2.1
Synthetic	DeepSeek V4 Flash (low, auto tool choice)	raw	192	0	0	94.3	98.4	94.3	98.4
Synthetic	DeepSeek V4 Flash (low, auto tool choice)	canonical	192	0	0	97.9	99.0	97.9	99.0
Synthetic	DeepSeek V4 Flash (low, auto tool choice)	snapshot	192	0	0	99.0	100.0	100.0	99.0
Synthetic	DeepSeek V4 Flash (low, auto tool choice)	state-raw	192	0	0	94.8	99.5	94.8	99.5
Synthetic	DeepSeek V4 Flash (low, auto tool choice)	state-canonical	192	1	0	97.4	99.0	97.4	99.0
Synthetic	DeepSeek V4 Flash (high, forced call)	raw	192	12	10	1.6	6.8	1.6	1.6
Synthetic	DeepSeek V4 Flash (high, forced call)	canonical	192	18	15	1.0	3.6	1.0	1.0
Synthetic	DeepSeek V4 Flash (high, auto tool choice)	raw	192	2	2	92.2	94.3	93.2	93.2
Synthetic	DeepSeek V4 Flash (high, auto tool choice)	canonical	192	0	0	96.4	97.9	96.9	97.4
Synthetic	GLM-5.3 Flash (low, forced call)	raw	192	0	0	38.0	78.1	57.8	52.6
Synthetic	GLM-5.3 Flash (low, forced call)	canonical	192	0	0	53.6	84.4	69.8	62.5
Synthetic	GLM-5.3 Flash (low, auto tool choice)	raw	192	0	0	43.8	76.0	59.4	55.7
Synthetic	GLM-5.3 Flash (low, auto tool choice)	canonical	192	2	2	50.5	85.9	67.7	64.1
Synthetic	GLM-5.3 Flash (low, auto tool choice)	snapshot	192	0	0	58.9	100.0	100.0	58.9
Synthetic	GLM-5.3 Flash (low, auto tool choice)	state-raw	192	0	0	52.6	77.6	52.6	77.6
Synthetic	GLM-5.3 Flash (low, auto tool choice)	state-canonical	192	0	0	69.3	87.0	69.3	87.0
Synthetic	GLM-5.3 Flash (high, forced call)	raw	192	0	0	62.5	91.1	74.5	74.5
Synthetic	GLM-5.3 Flash (high, forced call)	canonical	192	0	0	74.5	93.8	83.3	82.3
Synthetic	GLM-5.3 Flash (high, auto tool choice)	raw	192	1	1	66.7	92.2	79.2	76.6
Synthetic	GLM-5.3 Flash (high, auto tool choice)	canonical	192	0	0	74.5	95.8	83.3	84.4
Synthetic	DeepSeek V4 Pro (low, forced call)	raw	192	6	6	94.3	95.8	94.3	96.4
Synthetic	DeepSeek V4 Pro (low, forced call)	canonical	192	2	2	97.4	99.0	97.9	98.4
Synthetic	DeepSeek V4 Pro (low, auto tool choice)	raw	192	0	0	98.4	99.0	98.4	99.0
Synthetic	DeepSeek V4 Pro (low, auto tool choice)	canonical	192	0	0	97.4	100.0	98.4	99.0
Synthetic	DeepSeek V4 Pro (low, auto tool choice)	snapshot	192	0	0	99.0	100.0	100.0	99.0
Synthetic	DeepSeek V4 Pro (low, auto tool choice)	state-raw	192	2	2	97.9	98.4	97.9	98.4
Synthetic	DeepSeek V4 Pro (low, auto tool choice)	state-canonical	192	0	0	96.9	99.0	96.9	99.0
Synthetic	DeepSeek V4 Pro (high, forced call)	raw	192	16	16	91.7	91.7	91.7	91.7
Synthetic	DeepSeek V4 Pro (high, forced call)	canonical	192	32	32	80.2	83.3	81.2	82.3
Synthetic	DeepSeek V4 Pro (high, auto tool choice)	raw	192	12	12	92.7	93.8	93.2	93.2
Synthetic	DeepSeek V4 Pro (high, auto tool choice)	canonical	192	2	2	96.4	97.4	96.4	97.4
Retail profiles	GPT-5.4 mini (low)	raw	96	1	0	4.2	14.6	4.2	12.5
Retail profiles	GPT-5.4 mini (low)	canonical	96	0	0	40.6	90.6	74.0	46.9
Retail profiles	GPT-5.4 mini (low)	snapshot	96	0	0	89.6	100.0	100.0	89.6
Retail profiles	GPT-5.4 mini (low)	state-raw	96	0	0	3.1	17.7	3.1	17.7
Retail profiles	GPT-5.4 mini (low)	state-canonical	96	0	0	89.6	99.0	89.6	99.0
Retail profiles	GPT-5.4 mini (low)	tool-raw	96	0	0	87.5	89.6	87.5	89.6
Retail profiles	GPT-5.4 mini (low)	tool-canonical	96	0	0	97.9	100.0	97.9	100.0
Retail profiles	GPT-5.4 mini (medium)	raw	96	0	0	11.5	24.0	11.5	16.7
Retail profiles	GPT-5.4 mini (medium)	canonical	96	0	0	36.5	52.1	41.7	38.5
Retail profiles	GPT-5.4 mini (medium)	snapshot	96	0	0	94.8	100.0	100.0	94.8
Retail profiles	GPT-5.4 mini (high)	raw	96	1	0	13.5	30.2	13.5	20.8
Retail profiles	GPT-5.4 mini (high)	canonical	96	0	0	27.1	35.4	28.1	29.2
Retail profiles	GPT-5.4 mini (high)	snapshot	96	0	0	93.8	100.0	100.0	93.8
Retail profiles	GPT-5.4 mini (high)	state-raw	96	0	0	21.9	34.4	21.9	34.4
Retail profiles	GPT-5.4 mini (high)	state-canonical	96	0	0	26.0	34.4	26.0	34.4
Retail profiles	GPT-5.4 mini (low, auto tool choice)	raw	96	0	0	4.2	18.8	5.2	14.6
Retail profiles	GPT-5.4 mini (low, auto tool choice)	canonical	96	1	1	71.9	92.7	79.2	82.3
Retail profiles	GPT-5.4 mini (low, auto tool choice)	snapshot	96	0	0	89.6	100.0	100.0	89.6
Retail profiles	GPT-5.4 mini (low, auto tool choice)	state-raw	96	1	1	4.2	19.8	4.2	19.8
Retail profiles	GPT-5.4 mini (low, auto tool choice)	state-canonical	96	2	2	84.4	94.8	84.4	94.8
Retail profiles	GPT-5.4 mini (high, auto tool choice)	raw	96	1	1	35.4	49.0	35.4	49.0
Retail profiles	GPT-5.4 mini (high, auto tool choice)	canonical	96	1	1	93.8	96.9	95.8	94.8
Retail profiles	GPT-5.4 (low)	raw	96	0	0	88.5	90.6	88.5	90.6
Retail profiles	GPT-5.4 (low)	canonical	96	0	0	91.7	100.0	100.0	91.7
Retail profiles	GPT-5.4 (low)	snapshot	96	0	0	99.0	100.0	100.0	99.0
Retail profiles	GPT-5.4 (low)	state-raw	96	0	0	85.4	90.6	85.4	90.6
Retail profiles	GPT-5.4 (low)	state-canonical	96	0	0	99.0	99.0	99.0	99.0
Retail profiles	GPT-5.4 (low)	tool-raw	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	GPT-5.4 (low)	tool-canonical	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	GPT-5.4 (medium)	raw	96	0	0	95.8	95.8	95.8	95.8
Retail profiles	GPT-5.4 (medium)	canonical	96	0	0	94.8	99.0	99.0	94.8
Retail profiles	GPT-5.4 (high)	raw	96	0	0	99.0	99.0	99.0	99.0
Retail profiles	GPT-5.4 (high)	canonical	96	0	0	88.5	90.6	90.6	88.5
Retail profiles	GPT-5.4 (low, auto tool choice)	raw	96	0	0	95.8	97.9	95.8	97.9
Retail profiles	GPT-5.4 (low, auto tool choice)	canonical	96	0	0	100.0	100.0	100.0	100.0

continued on next page

Table 6 (continued)

Cohort	Configuration	Arm	N	Invalid	Missing	All five	Stock	Hist. joint	Future joint
Retail profiles	GPT-5.4 (low, auto tool choice)	snapshot	96	0	0	89.6	100.0	100.0	89.6
Retail profiles	GPT-5.4 (low, auto tool choice)	state-raw	96	0	0	93.8	95.8	93.8	95.8
Retail profiles	GPT-5.4 (low, auto tool choice)	state-canonical	96	0	0	91.7	91.7	91.7	91.7
Retail profiles	GPT-5.4 (high, auto tool choice)	raw	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	GPT-5.4 (high, auto tool choice)	canonical	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	Haiku 4.5	raw	96	0	0	1.0	7.3	1.0	6.2
Retail profiles	Haiku 4.5	canonical	96	0	0	71.9	96.9	77.1	89.6
Retail profiles	Haiku 4.5	snapshot	96	0	0	92.7	100.0	100.0	92.7
Retail profiles	Haiku 4.5	state-raw	96	0	0	1.0	6.2	1.0	6.2
Retail profiles	Haiku 4.5	state-canonical	96	0	0	78.1	94.8	78.1	94.8
Retail profiles	Haiku 4.5 + thinking	raw	96	37	37	33.3	41.7	34.4	40.6
Retail profiles	Haiku 4.5 + thinking	canonical	96	0	0	87.5	99.0	93.8	91.7
Retail profiles	Haiku 4.5 + thinking	snapshot	96	0	0	96.9	100.0	100.0	96.9
Retail profiles	Haiku 4.5 + thinking	state-raw	96	22	21	46.9	59.4	46.9	59.4
Retail profiles	Haiku 4.5 + thinking	state-canonical	96	0	0	94.8	99.0	94.8	99.0
Retail profiles	Sonnet 4.6	raw	96	0	0	91.7	96.9	91.7	95.8
Retail profiles	Sonnet 4.6	canonical	96	0	0	97.9	100.0	99.0	99.0
Retail profiles	Sonnet 4.6	snapshot	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	Sonnet 4.6	state-raw	96	0	0	92.7	95.8	92.7	95.8
Retail profiles	Sonnet 4.6	state-canonical	96	0	0	99.0	100.0	99.0	100.0
Retail profiles	Sonnet + thinking	raw	96	0	0	96.9	99.0	97.9	97.9
Retail profiles	Sonnet + thinking	canonical	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	Sonnet + thinking	snapshot	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	Sonnet + thinking	state-raw	96	0	0	96.9	100.0	96.9	100.0
Retail profiles	Sonnet + thinking	state-canonical	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	gpt-oss-20b (low, forced call)	raw	96	73	73	0.0	3.1	0.0	1.0
Retail profiles	gpt-oss-20b (low, forced call)	canonical	96	53	53	7.3	17.7	9.4	11.5
Retail profiles	gpt-oss-20b (low, auto tool choice)	raw	96	69	69	0.0	2.1	0.0	0.0
Retail profiles	gpt-oss-20b (low, auto tool choice)	canonical	96	58	58	14.6	19.8	16.7	17.7
Retail profiles	gpt-oss-20b (low, auto tool choice)	snapshot	96	1	1	77.1	97.9	97.9	77.1
Retail profiles	gpt-oss-20b (high, forced call)	raw	96	96	96	0.0	0.0	0.0	0.0
Retail profiles	gpt-oss-20b (high, forced call)	canonical	96	49	49	45.8	47.9	45.8	47.9
Retail profiles	gpt-oss-20b (high, auto tool choice)	raw	96	96	96	0.0	0.0	0.0	0.0
Retail profiles	gpt-oss-20b (high, auto tool choice)	canonical	96	50	50	46.9	47.9	47.9	46.9
Retail profiles	gpt-oss-120b (low, forced call)	raw	96	0	0	1.0	17.7	3.1	7.3
Retail profiles	gpt-oss-120b (low, forced call)	canonical	96	0	0	69.8	85.4	78.1	76.0
Retail profiles	gpt-oss-120b (low, auto tool choice)	raw	96	47	47	2.1	10.4	2.1	5.2
Retail profiles	gpt-oss-120b (low, auto tool choice)	canonical	96	1	1	70.8	83.3	79.2	72.9
Retail profiles	gpt-oss-120b (low, auto tool choice)	snapshot	96	1	0	86.5	97.9	97.9	86.5
Retail profiles	gpt-oss-120b (high, forced call)	raw	96	96	96	0.0	0.0	0.0	0.0
Retail profiles	gpt-oss-120b (high, forced call)	canonical	96	85	85	11.5	11.5	11.5	11.5
Retail profiles	gpt-oss-120b (high, auto tool choice)	raw	96	96	96	0.0	0.0	0.0	0.0
Retail profiles	gpt-oss-120b (high, auto tool choice)	canonical	96	86	86	10.4	10.4	10.4	10.4
Retail profiles	Qwen3.8-27B (low, auto tool choice)	raw	96	0	0	93.8	96.9	93.8	96.9
Retail profiles	Qwen3.8-27B (low, auto tool choice)	canonical	96	0	0	99.0	100.0	100.0	99.0
Retail profiles	Qwen3.8-27B (low, auto tool choice)	snapshot	96	0	0	100.0	100.0	100.0	100.0
Retail profiles	Qwen3.8-27B (high, auto tool choice)	raw	96	0	0	96.9	97.9	96.9	97.9
Retail profiles	Qwen3.8-27B (high, auto tool choice)	canonical	96	0	0	95.8	99.0	97.9	96.9
Retail profiles	DeepSeek V4 Flash (low, forced call)	raw	96	0	0	0.0	7.3	1.0	0.0
Retail profiles	DeepSeek V4 Flash (low, forced call)	canonical	96	7	4	0.0	10.4	0.0	1.0
Retail profiles	DeepSeek V4 Flash (low, auto tool choice)	raw	96	2	1	91.7	94.8	91.7	94.8
Retail profiles	DeepSeek V4 Flash (low, auto tool choice)	canonical	96	1	1	95.8	97.9	96.9	96.9
Retail profiles	DeepSeek V4 Flash (low, auto tool choice)	snapshot	96	0	0	99.0	100.0	100.0	99.0
Retail profiles	DeepSeek V4 Flash (high, forced call)	raw	96	11	8	0.0	5.2	2.1	1.0
Retail profiles	DeepSeek V4 Flash (high, forced call)	canonical	96	1	1	0.0	10.4	0.0	1.0
Retail profiles	DeepSeek V4 Flash (high, auto tool choice)	raw	96	3	3	93.8	94.8	93.8	94.8
Retail profiles	DeepSeek V4 Flash (high, auto tool choice)	canonical	96	1	0	99.0	99.0	99.0	99.0
Retail profiles	GLM-5.3 Flash (low, forced call)	raw	96	0	0	22.9	53.1	28.1	31.2
Retail profiles	GLM-5.3 Flash (low, forced call)	canonical	96	0	0	49.0	90.6	67.7	64.6
Retail profiles	GLM-5.3 Flash (low, auto tool choice)	raw	96	0	0	11.5	51.0	21.9	26.0
Retail profiles	GLM-5.3 Flash (low, auto tool choice)	canonical	96	0	0	43.8	87.5	66.7	58.3
Retail profiles	GLM-5.3 Flash (low, auto tool choice)	snapshot	96	0	0	38.5	100.0	100.0	38.5
Retail profiles	GLM-5.3 Flash (high, forced call)	raw	96	0	0	59.4	87.5	71.9	70.8
Retail profiles	GLM-5.3 Flash (high, forced call)	canonical	96	0	0	66.7	94.8	75.0	83.3
Retail profiles	GLM-5.3 Flash (high, auto tool choice)	raw	96	0	0	64.6	84.4	75.0	70.8
Retail profiles	GLM-5.3 Flash (high, auto tool choice)	canonical	96	0	0	68.8	94.8	76.0	84.4
Retail profiles	DeepSeek V4 Pro (low, forced call)	raw	96	7	7	90.6	91.7	90.6	91.7
Retail profiles	DeepSeek V4 Pro (low, forced call)	canonical	96	2	2	95.8	97.9	96.9	96.9
Retail profiles	DeepSeek V4 Pro (low, auto tool choice)	raw	96	0	0	94.8	97.9	96.9	95.8
Retail profiles	DeepSeek V4 Pro (low, auto tool choice)	canonical	96	0	0	99.0	100.0	99.0	100.0
Retail profiles	DeepSeek V4 Pro (low, auto tool choice)	snapshot	96	0	0	95.8	100.0	100.0	95.8
Retail profiles	DeepSeek V4 Pro (high, forced call)	raw	96	19	19	79.2	79.2	79.2	79.2
Retail profiles	DeepSeek V4 Pro (high, forced call)	canonical	96	20	20	78.1	78.1	78.1	78.1
Retail profiles	DeepSeek V4 Pro (high, auto tool choice)	raw	96	3	3	95.8	96.9	95.8	96.9
Retail profiles	DeepSeek V4 Pro (high, auto tool choice)	canonical	96	0	0	99.0	100.0	99.0	100.0

Configuration	K-chg	K-both	D-same	D-both
GPT-5.4 mini (low)	97.1	8.8	27.1	25.0
GPT-5.4 (low)	100.0	73.5	68.8	68.8
Sonnet 4.6	100.0	88.2	89.6	89.6
Sonnet + thinking	100.0	97.1	93.8	93.8

Table 7: Paired raw controls (%). K-chg/K-both: changed answers/both correct on 17 bases with changed truth. D-same/D-both: equal valid answers/both correct on 24 display-permuted bases. Rates average repetitions within base. Change alone does not establish correct sensitivity.

C Paired temporal and display controls

On the first 24 heldout bases (one factorial cycle), a paired view changes only K to 130, revealing later reports. This is retrospective revised-state simulation, not a forecast made at time 100. Another control shuffles raw report display order while preserving all timestamps and the correct answer. We report both-view correctness as well as sensitivity and invariance, since consistent wrong answers are not evidence of reliability (Table 7). Separate requests also vary stochastically; cross-view disagreement alone does not isolate an order effect. Table 8 uses the existing two repetitions on the same 24 control bases. Within-view pairs compare repetitions 0 and 1; cross-view pairs compare corresponding repetitions across the original and shuffled views. All are separate API requests, with no shared random seed. Thus cross-view disagreement includes ordinary request variation. Mini's cross-view agreement of 27.1% lies between its within-view rates of 25.0% and 33.3%; these observations do not isolate a causal display-order effect. This diagnostic was added after collection of the synthetic cohorts, without additional model calls.

Configuration	Original repeats	Shuffled repeats	Cross-view
GPT-5.4 mini (low)	25.0 / 20.8	33.3 / 33.3	27.1 / 25.0
GPT-5.4 (low)	70.8 / 70.8	70.8 / 70.8	68.8 / 68.8
Sonnet 4.6	95.8 / 95.8	83.3 / 83.3	89.6 / 89.6
Sonnet + thinking	91.7 / 91.7	95.8 / 95.8	93.8 / 93.8

Table 8: Post-hoc repeat diagnostics (%): identical valid answers / both correct. Within-view columns have 24 repetition pairs; cross-view has 48 matched-repetition pairs over those same 24 bases. These are descriptive rates, not independent samples of histories.

D Consistency and decision diagnostics

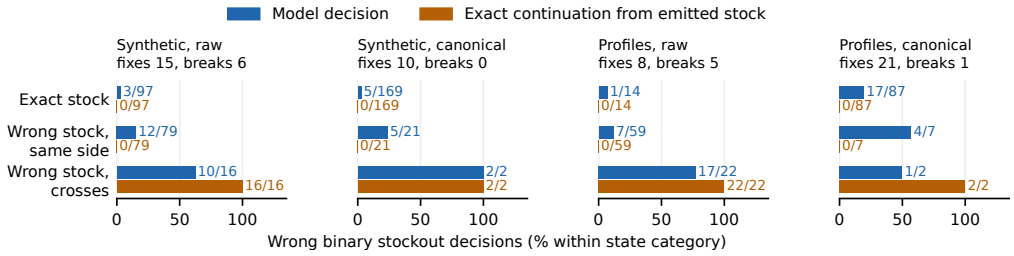


Figure 1: GPT-5.4 mini (low): wrong binary stockout decisions within each state category, for the model's decision and for exact continuation from its emitted stock ("fixes": wrong model decisions made right by continuation; "breaks": right decisions made wrong). Continuation is error-free by construction in the first two rows and always wrong in the third; the empirical content is the category sizes and the model decisions. The same-side share depends on how far true stocks sit from the threshold: median distance 43 units on synthetic histories and 11 on profiles. Descriptive percentages.

Primary synthetic: among pooled raw and canonical requests, 70/139 valid outputs with both wrong starting stock and wrong future quantities exactly match replay from the emitted stock. There are 60 correct-stock/wrong-future outputs and 0 wrong-stock/correct-future outputs among 1536 valid

requests. Historical loss is wrong despite correct starting stock in 71 requests. Across all arms, 0/2304 contracts are invalid and 0 responses hit the output-token limit (0 overlap).

Later knowledge: among pooled raw and canonical requests, 18/32 valid outputs with both wrong starting stock and wrong future quantities exactly match replay from the emitted stock. There are 15 correct-stock/wrong-future outputs and 0 wrong-stock/correct-future outputs among 384 valid requests. Historical loss is wrong despite correct starting stock in 17 requests. Across all arms, 0/576 contracts are invalid and 0 responses hit the output-token limit (0 overlap).

Display permutation: among raw requests, 16/35 valid outputs with both wrong starting stock and wrong future quantities exactly match replay from the emitted stock. There are 0 correct-stock/wrong-future outputs and 0 wrong-stock/correct-future outputs among 192 valid requests. Historical loss is wrong despite correct starting stock in 8 requests. Across all arms, 0/192 contracts are invalid and 0 responses hit the output-token limit (0 overlap).

Retail profiles: among pooled raw and canonical requests, 57/102 valid outputs with both wrong starting stock and wrong future quantities exactly match replay from the emitted stock. There are 56 correct-stock/wrong-future outputs and 1 wrong-stock/correct-future outputs among 767 valid requests. Historical loss is wrong despite correct starting stock in 35 requests. Across all arms, 1/1152 contracts are invalid and 0 responses hit the output-token limit (0 overlap).

These are descriptive counts over repeated requests, not independent failure probabilities. Replay agreement establishes observable consistency, not that the model internally used its emitted stock; correct future outputs after a wrong stock do not prove intentional recovery. Invalid and truncated counts overlap and must not be added.

An observed failure with a scenario consequence. As a post-hoc illustration, primary base `b0007`, mini raw repetition 0, returns starting stock 31 instead of 18. Its future outputs are terminal stock 6, zero loss, and no stockout. Exact rollout from its emitted stock of 31 produces precisely those outputs, while rollout from the correct stock of 18 ends at zero, loses seven units, and first stocks out at time 130. This trace is consistent with a wrong-state rollout, without establishing the model's internal process. It is illustrative rather than a separately sampled test. Both repetitions of this request return the identical five fields, so repeat agreement would not have flagged the error.

For any consistent rollout, terminal stock minus future loss equals starting stock plus supplied receipts minus supplied demand. Consequently, the correct terminal-stock/loss pair cannot be obtained by consistent replay from a wrong starting stock in this environment. An output with wrong stock and correct future quantities would demonstrate observable inconsistency, not reliable recovery. Historical-loss errors can still coexist with correct starting stock and future outputs.

Descriptive task-length strata. Mini raw all-five success is 39/64, 16/64, and 8/64 for 12, 36, and 72 historical events, respectively; future-joint success is 43/64, 18/64, and 11/64. These are different seeded histories within the factorial design, not matched interventions identifying an isolated length or temporal-reasoning effect. All configuration and factor strata remain available in the analysis artifact.

Why correct totals do not certify replay. A post-hoc snapshot trace (mini, primary `b0019`, repetition 0) copies starting stock 9 correctly but emits final stock 0 and future loss 79 instead of 8 and 87. Both pairs satisfy the same conservation equation. The final receipt adds eight units after the last demand; lost sales cannot be filled retroactively, so these eight units must remain. This shows why event-order execution remains necessary after correct historical reconstruction, without inferring the model's internal algorithm.

Executable continuation (post-hoc). Table 9 retains all input arms and configurations for the two main cohorts; the supplement also includes both control cohorts. This is a deterministic transformation of original answers, not newly sampled model behavior. The original prompt requests all five fields, and the strict five-field validity gate is retained even if an otherwise invalid answer has a usable stock field. For valid answers, continuation future-joint accuracy equals starting-stock accuracy: conservation makes correct final stock and loss impossible under consistent replay from an incorrect stock. This equivalence is an endpoint property, not a new inventory algorithm. An accidentally

correct future triple paired with wrong stock can regress; this occurs once for mini's raw retail responses. No such case is removed.

The operational false-alarm/miss endpoints need not move together. For mini's canonical synthetic responses, execution reduces false alarms from 5/70 to 1/70 and misses from 7/122 to 1/122; canonical retail rates improve from 15/46 to 1/46 and 7/50 to 1/50. For raw retail responses, false alarms rise from 15/46 to 19/46 while misses fall from 10/49 to 3/49. These rates condition on valid answers; the invalid raw retail response belongs to a positive-stockout task and remains a failure in the unconditional accuracy. They measure detection, not timing.

Configuration	Input	Synthetic	Retail profiles
GPT-5.4 mini (low)	raw	37.5 / 50.5	12.5 / 14.6
GPT-5.4 mini (low)	canonical	76.0 / 88.0	46.9 / 90.6
GPT-5.4 mini (low)	snapshot	89.6 / 100.0	89.6 / 100.0
GPT-5.4 (low)	raw	88.0 / 91.1	90.6 / 90.6
GPT-5.4 (low)	canonical	98.4 / 99.5	91.7 / 100.0
GPT-5.4 (low)	snapshot	99.0 / 100.0	99.0 / 100.0
Sonnet 4.6	raw	99.0 / 99.5	95.8 / 96.9
Sonnet 4.6	canonical	99.5 / 100.0	99.0 / 100.0
Sonnet 4.6	snapshot	100.0 / 100.0	100.0 / 100.0
Sonnet + thinking	raw	97.9 / 99.0	97.9 / 99.0
Sonnet + thinking	canonical	100.0 / 100.0	100.0 / 100.0
Sonnet + thinking	snapshot	100.0 / 100.0	100.0 / 100.0

Table 9: Post-hoc future-joint accuracy (%): original model rollout / executable continuation from its emitted stock. Synthetic and profile cells contain 192 and 96 original calls, respectively. Invalid five-field answers fail both endpoints. No state-only prompts were run; full-code replay is exact on every task.

E Extension studies: protocol summary and full results

Both extension studies were specified in a written protocol after the original outcomes were known and before any of their calls, with stated hypotheses, and are reported in full whatever they show. They reuse the archived heldout histories, so they add configurations and arms rather than independent histories. The effort sweeps hold every prompt, task and schema fixed and change only the provider's requested reasoning effort (GPT-5.4 mini at medium and high on all arms; GPT-5.4 at medium and high on raw and canonical input, added by Amendment 2). The code-inside arms add the provider's hosted code-execution tool to the two OpenAI configurations with automatic tool choice, so the model decides whether to run code; the instructions gain one sentence saying the tool is available. The state-only arms withhold the future schedule and horizon, ask for the two historical integers, and apply the schedule in code; invalid two-field answers fail and are not repaired. A same-period replication of the original low-effort synthetic cells was added by amendment, before its calls, to rule out a collection-date confound in the effort comparison. Every response was reparsed and rescored independently of the runner, the planned grids were verified complete, task payloads were checked byte-for-byte against the archived cohorts, and the conservation identity was verified on every hybrid answer. Full per-cell counts, paired differences with intervals, token usage and cost are in the supplement's extension audit; Table 6 lists every cell and Tables 10 and 11 every paired difference. The anonymous supplement contains all code, generators, prompts, raw provider responses, audits and the reasoning-summary traces of diagnostic D1.

Same-period low-effort replication. The 576-call replication of mini's low-effort synthetic cells gives original versus replication future-joint accuracy of raw 37.5 vs 35.9% (-1.6 [-9.4, +5.7] points), canonical 76.0 vs 67.2% (-8.9 [-16.7, -1.0] points), snapshot 89.6 vs 89.1% (-0.5 [-6.2, +5.2] points), and starting-stock accuracy of raw 50.5 vs 49.0%, canonical 88.0 vs 84.4%. The canonical cell replicates lower than the original, so collection-period variation of that size should be assumed when reading any single mini cell. Paired changes of the raised-effort cells relative to the replication baseline (future-joint, percentage points): GPT-5.4 mini (medium)/raw -9.4 [-17.2, -1.0]; GPT-5.4 mini (medium)/canonical -20.3 [-28.6, -12.0]; GPT-5.4 mini (medium)/snapshot +8.9 [+3.6, +14.6];

GPT-5.4 mini (high)/raw -10.9 [-19.8, -1.6]; GPT-5.4 mini (high)/canonical -32.3 [-42.2, -22.4]; GPT-5.4 mini (high)/snapshot +8.3 [+3.1, +14.1]. Paired changes of the state-only arms relative to the replication baseline: raw stock exact +6.2 [-1.0, +13.5]; raw future joint +19.3 [+10.9, +27.6]; canonical stock exact +9.4 [+3.6, +15.6]; canonical future joint +26.6 [+19.3, +34.4].

Character of the raised-effort canonical errors. Among mini's wrong canonical starting stocks at high effort, 79 of 83 synthetic and 49 of 62 profile answers are underestimates, with median absolute errors of 32 and 14 units; 57 and 41 are exactly zero although the true stock is positive, 57 and 41 equal the stock obtained by replaying only the demands, and 13 and 13 equal the clipped net total. In twelve re-issued requests with reasoning summaries (diagnostic D1, supplement), the summaries lay out the correct selection and replay procedure, yet several answers are still zero or far too low, and one summary states that it must call the answer tool immediately and considers submitting placeholder zeros. This suggests that, under forced tool choice at raised effort, the model sometimes emits an answer before completing the replay it planned; the traces are illustrative, not a measured mechanism.

Code-inside arm for mini on synthetic histories, by history length. tool-raw: 12 events 52 of 64 without code (43 exact), 0 wrong with code, 36 events 1 of 64 without code (0 exact), 2 wrong with code, 72 events 0 of 64 without code (0 exact), 4 wrong with code; tool-canonical: 12 events 51 of 64 without code (47 exact), 0 wrong with code, 36 events 2 of 64 without code (2 exact), 0 wrong with code, 72 events 0 of 64 without code (0 exact), 0 wrong with code. Calls answered without code concentrate on the shortest histories; the answers that were wrong despite running code are counted per stratum and were not individually inspected. The Anthropic configurations were not run with a code tool because their tool interface differs and both are at ceiling without it.

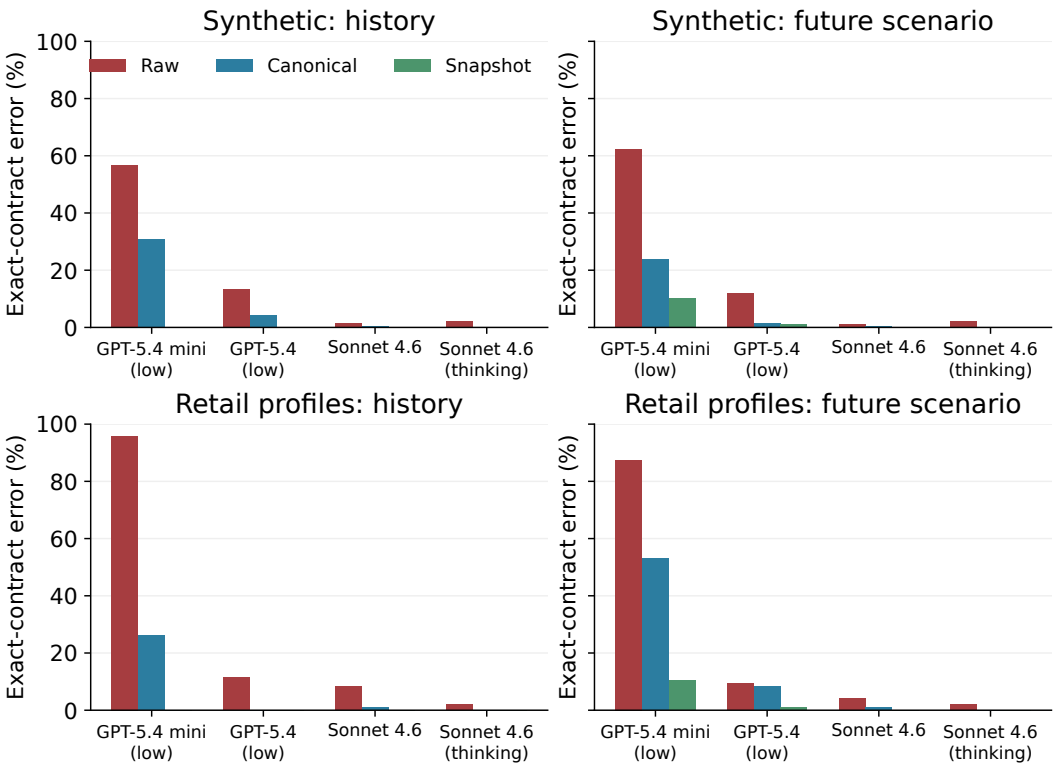


Figure 2: Historical and future-stage errors under the three representations in the original grid. Snapshot historical answers are supplied by code; their accuracy measures faithful copying. Future-stage errors remain the substantive endpoint for that representation.

Comparison (right minus left)	Metric	Bases	Difference	95% interval
GPT-5.4 mini (low)/raw → canonical	future joint	96	+38.5	[+28.6, +47.9]
GPT-5.4 mini (low)/raw → canonical	stock exact	96	+37.5	[+28.6, +46.4]
GPT-5.4 mini (low)/raw → snapshot	future joint	96	+52.1	[+43.8, +60.4]
GPT-5.4 mini (low)/raw → snapshot	stock exact	96	+49.5	[+41.1, +58.3]
GPT-5.4 (low)/raw → canonical	future joint	96	+10.4	[+5.7, +15.6]
GPT-5.4 (low)/raw → canonical	stock exact	96	+8.3	[+3.6, +13.5]
GPT-5.4 (low)/raw → snapshot	future joint	96	+10.9	[+6.2, +16.7]
GPT-5.4 (low)/raw → snapshot	stock exact	96	+8.9	[+4.7, +14.1]
Sonnet 4.6/raw → canonical	future joint	96	+0.5	[-1.0, +2.6]
Sonnet 4.6/raw → canonical	stock exact	96	+0.5	[+0.0, +1.6]
Sonnet 4.6/raw → snapshot	future joint	96	+1.0	[+0.0, +2.6]
Sonnet 4.6/raw → snapshot	stock exact	96	+0.5	[+0.0, +1.6]
Sonnet + thinking/raw → canonical	future joint	96	+2.1	[+0.5, +4.2]
Sonnet + thinking/raw → canonical	stock exact	96	+1.0	[+0.0, +2.6]
Sonnet + thinking/raw → snapshot	future joint	96	+2.1	[+0.5, +4.2]
Sonnet + thinking/raw → snapshot	stock exact	96	+1.0	[+0.0, +2.6]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (medium)/raw	future joint	96	-10.9	[-18.8, -3.1]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (medium)/raw	stock exact	96	-4.7	[-13.0, +3.6]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (medium)/canonical	future joint	96	-29.2	[-38.0, -20.3]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (medium)/canonical	stock exact	96	-21.9	[-29.2, -14.6]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (medium)/snapshot	future joint	96	+8.3	[+4.2, +12.5]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (medium)/snapshot	stock exact	96	+0.0	[+0.0, +0.0]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (high)/raw	future joint	96	-12.5	[-21.9, -3.1]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (high)/raw	stock exact	96	-8.9	[-17.7, +0.5]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (high)/canonical	future joint	96	-41.1	[-51.0, -31.2]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (high)/canonical	stock exact	96	-31.8	[-40.6, -22.9]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (high)/snapshot	future joint	96	+7.8	[+3.1, +12.5]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (high)/snapshot	stock exact	96	+0.0	[+0.0, +0.0]
GPT-5.4 (low)/raw → GPT-5.4 (medium)/raw	future joint	96	+7.3	[+1.0, +14.1]
GPT-5.4 (low)/raw → GPT-5.4 (medium)/raw	stock exact	96	+7.3	[+2.6, +12.5]
GPT-5.4 (low)/canonical → GPT-5.4 (medium)/canonical	future joint	96	-2.6	[-5.7, +0.5]
GPT-5.4 (low)/canonical → GPT-5.4 (medium)/canonical	stock exact	96	-1.6	[-4.2, +0.5]
GPT-5.4 (low)/raw → GPT-5.4 (high)/raw	future joint	96	+6.2	[+0.0, +13.0]
GPT-5.4 (low)/raw → GPT-5.4 (high)/raw	stock exact	96	+5.7	[+0.5, +11.5]
GPT-5.4 (low)/canonical → GPT-5.4 (high)/canonical	future joint	96	-3.6	[-7.8, +0.0]
GPT-5.4 (low)/canonical → GPT-5.4 (high)/canonical	stock exact	96	-1.0	[-3.1, +1.0]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (high, auto tool choice)/raw	future joint	96	+22.4	[+14.1, +30.7]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (low, auto tool choice)/raw	future joint	96	+19.3	[+11.5, +27.1]
GPT-5.4 mini (high)/raw → GPT-5.4 mini (high, auto tool choice)/raw	future joint	96	+54.2	[+44.3, +63.0]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (high, auto tool choice)/raw	stock exact	96	+23.4	[+15.1, +31.2]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (low, auto tool choice)/raw	stock exact	96	+8.3	[+2.1, +15.1]
GPT-5.4 mini (high)/raw → GPT-5.4 mini (high, auto tool choice)/raw	stock exact	96	+40.6	[+31.2, +49.5]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	future joint	96	+6.8	[+0.5, +13.5]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (low, auto tool choice)/canonical	future joint	96	+10.4	[+3.1, +18.2]
GPT-5.4 mini (high)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	future joint	96	+58.3	[+50.0, +66.7]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	stock exact	96	+5.2	[+0.5, +10.4]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (low, auto tool choice)/canonical	stock exact	96	+3.6	[-1.6, +8.9]
GPT-5.4 mini (high)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	stock exact	96	+40.6	[+31.8, +49.5]
GPT-5.4 mini (low)/raw → state-raw	stock exact	96	+4.7	[-2.1, +11.5]
GPT-5.4 mini (low)/raw → state-raw	future joint	96	+17.7	[+9.9, +25.5]
GPT-5.4 mini (low)/canonical → state-canonical	stock exact	96	+5.7	[+0.0, +11.5]
GPT-5.4 mini (low)/canonical → state-canonical	future joint	96	+17.7	[+9.9, +25.5]
GPT-5.4 mini (high)/raw → state-raw	stock exact	96	+12.5	[+3.6, +20.8]
GPT-5.4 mini (high)/raw → state-raw	future joint	96	+29.2	[+19.8, +38.5]
GPT-5.4 mini (high)/canonical → state-canonical	stock exact	96	+15.1	[+7.3, +22.9]
GPT-5.4 mini (high)/canonical → state-canonical	future joint	96	+36.5	[+26.6, +46.4]
GPT-5.4 (low)/raw → state-raw	stock exact	96	+0.5	[-5.2, +6.2]
GPT-5.4 (low)/raw → state-raw	future joint	96	+3.6	[-2.6, +9.9]
GPT-5.4 (low)/canonical → state-canonical	stock exact	96	-1.0	[-3.1, +1.0]
GPT-5.4 (low)/canonical → state-canonical	future joint	96	+0.0	[-2.6, +2.6]
Sonnet 4.6/raw → state-raw	stock exact	96	+0.5	[+0.0, +1.6]
Sonnet 4.6/raw → state-raw	future joint	96	+1.0	[+0.0, +2.6]
Sonnet 4.6/canonical → state-canonical	stock exact	96	+0.0	[+0.0, +0.0]
Sonnet 4.6/canonical → state-canonical	future joint	96	+0.5	[+0.0, +1.6]
Sonnet + thinking/raw → state-raw	stock exact	96	+1.0	[+0.0, +2.6]
Sonnet + thinking/raw → state-raw	future joint	96	+2.1	[+0.5, +4.2]
Sonnet + thinking/canonical → state-canonical	stock exact	96	+0.0	[+0.0, +0.0]
Sonnet + thinking/canonical → state-canonical	future joint	96	+0.0	[+0.0, +0.0]
GPT-5.4 mini (low)/raw → tool-raw	future joint	96	+54.2	[+45.3, +62.5]
GPT-5.4 mini (low)/raw → tool-raw	stock exact	96	+44.3	[+35.4, +53.1]
GPT-5.4 mini (low)/state-raw → tool-raw	future joint	96	+36.5	[+27.1, +45.8]
GPT-5.4 mini (low)/canonical → tool-canonical	future joint	96	+21.9	[+14.6, +29.2]
GPT-5.4 mini (low)/canonical → tool-canonical	stock exact	96	+11.5	[+6.2, +16.7]
GPT-5.4 mini (low)/state-canonical → tool-canonical	future joint	96	+4.2	[+0.5, +8.3]
GPT-5.4 (low)/raw → tool-raw	future joint	96	+12.0	[+7.3, +17.2]
GPT-5.4 (low)/raw → tool-raw	stock exact	96	+8.9	[+4.7, +14.1]
GPT-5.4 (low)/state-raw → tool-raw	future joint	96	+8.3	[+4.7, +12.5]
GPT-5.4 (low)/canonical → tool-canonical	future joint	96	-1.0	[-4.2, +1.6]
GPT-5.4 (low)/canonical → tool-canonical	stock exact	96	-1.0	[-3.1, +1.0]
GPT-5.4 (low)/state-canonical → tool-canonical	future joint	96	-1.0	[-4.2, +1.6]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (low)/tool-raw	future joint	96	+34.9	[+26.0, +43.8]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (low)/tool-raw	stock exact	96	+35.9	[+27.1, +44.8]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (low)/tool-canonical	future joint	96	+11.5	[+6.2, +17.7]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (low)/tool-canonical	stock exact	96	+7.8	[+3.6, +12.5]

Table 10: Paired differences in percentage points, synthetic cohort: matched by history and repetition index, averaged within history; descriptive 95% history-bootstrap percentile intervals (10,000 draws), not multiplicity adjusted. The same-period replication moved one identical cell by -8.9 points, so differences of that size are within run-to-run variation. Cells at or near 100% give degenerate intervals. Identical rows arise when two comparisons have identical per-history differences.

Comparison (right minus left)	Metric	Bases	Difference	95% interval
GPT-5.4 mini (low)/raw → canonical	future joint	48	+34.4	[+24.0, +44.8]
GPT-5.4 mini (low)/raw → canonical	stock exact	48	+76.0	[+66.7, +84.4]
GPT-5.4 mini (low)/raw → snapshot	future joint	48	+77.1	[+68.8, +85.4]
GPT-5.4 mini (low)/raw → snapshot	stock exact	48	+85.4	[+78.1, +91.7]
GPT-5.4 (low)/raw → canonical	future joint	48	+1.0	[-6.2, +8.3]
GPT-5.4 (low)/raw → canonical	stock exact	48	+9.4	[+4.2, +15.6]
GPT-5.4 (low)/raw → snapshot	future joint	48	+8.3	[+3.1, +14.6]
GPT-5.4 (low)/raw → snapshot	stock exact	48	+9.4	[+4.2, +15.6]
Sonnet 4.6/raw → canonical	future joint	48	+3.1	[+0.0, +8.3]
Sonnet 4.6/raw → canonical	stock exact	48	+3.1	[+0.0, +8.3]
Sonnet 4.6/raw → snapshot	future joint	48	+4.2	[+0.0, +9.4]
Sonnet 4.6/raw → snapshot	stock exact	48	+3.1	[+0.0, +8.3]
Sonnet + thinking/raw → canonical	future joint	48	+2.1	[+0.0, +6.2]
Sonnet + thinking/raw → canonical	stock exact	48	+1.0	[+0.0, +3.1]
Sonnet + thinking/raw → snapshot	future joint	48	+2.1	[+0.0, +6.2]
Sonnet + thinking/raw → snapshot	stock exact	48	+1.0	[+0.0, +3.1]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (medium)/raw	future joint	48	+4.2	[-4.2, +12.5]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (medium)/raw	stock exact	48	+9.4	[+1.0, +18.8]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (medium)/canonical	future joint	48	-8.3	[-20.8, +4.2]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (medium)/canonical	stock exact	48	-38.5	[-50.0, -27.1]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (medium)/snapshot	future joint	48	+5.2	[-1.0, +11.5]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (medium)/snapshot	stock exact	48	+0.0	[+0.0, +0.0]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (high)/raw	future joint	48	+8.3	[-1.0, +17.7]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (high)/raw	stock exact	48	+15.6	[+6.2, +25.0]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (high)/canonical	future joint	48	-17.7	[-30.2, -5.2]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (high)/canonical	stock exact	48	-55.2	[-65.6, -44.8]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (high)/snapshot	future joint	48	+4.2	[-4.2, +12.5]
GPT-5.4 mini (low)/snapshot → GPT-5.4 mini (high)/snapshot	stock exact	48	+0.0	[+0.0, +0.0]
GPT-5.4 (low)/raw → GPT-5.4 (medium)/raw	future joint	48	+5.2	[-2.1, +12.5]
GPT-5.4 (low)/raw → GPT-5.4 (medium)/raw	stock exact	48	+5.2	[-2.1, +12.5]
GPT-5.4 (low)/canonical → GPT-5.4 (medium)/canonical	future joint	48	+3.1	[-4.2, +10.4]
GPT-5.4 (low)/canonical → GPT-5.4 (medium)/canonical	stock exact	48	-1.0	[-3.1, +0.0]
GPT-5.4 (low)/raw → GPT-5.4 (high)/raw	future joint	48	+8.3	[+2.1, +14.6]
GPT-5.4 (low)/raw → GPT-5.4 (high)/raw	stock exact	48	+8.3	[+2.1, +14.6]
GPT-5.4 (low)/canonical → GPT-5.4 (high)/canonical	future joint	48	-3.1	[-11.5, +4.2]
GPT-5.4 (low)/canonical → GPT-5.4 (high)/canonical	stock exact	48	-9.4	[-15.6, -4.2]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (high, auto tool choice)/raw	future joint	48	+34.4	[+21.9, +46.9]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (low, auto tool choice)/raw	future joint	48	+2.1	[-6.2, +10.4]
GPT-5.4 mini (high)/raw → GPT-5.4 mini (high, auto tool choice)/raw	future joint	48	+28.1	[+16.7, +39.6]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (high, auto tool choice)/raw	stock exact	48	+30.2	[+18.8, +41.7]
GPT-5.4 mini (low)/raw → GPT-5.4 mini (low, auto tool choice)/raw	stock exact	48	+4.2	[-5.2, +13.5]
GPT-5.4 mini (high)/raw → GPT-5.4 mini (high, auto tool choice)/raw	stock exact	48	+18.8	[+7.3, +30.2]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	future joint	48	+12.5	[+4.2, +19.8]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (low, auto tool choice)/canonical	future joint	48	+35.4	[+24.0, +46.9]
GPT-5.4 mini (high)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	future joint	48	+65.6	[+56.2, +75.0]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	stock exact	48	+4.2	[-2.1, +10.4]
GPT-5.4 mini (low)/canonical → GPT-5.4 mini (low, auto tool choice)/canonical	stock exact	48	+2.1	[-6.2, +11.5]
GPT-5.4 mini (high)/canonical → GPT-5.4 mini (high, auto tool choice)/canonical	stock exact	48	+61.5	[+51.0, +71.9]
GPT-5.4 mini (low)/raw → state-raw	stock exact	48	+3.1	[-5.2, +11.5]
GPT-5.4 mini (low)/raw → state-raw	future joint	48	+5.2	[-3.1, +13.5]
GPT-5.4 mini (low)/canonical → state-canonical	stock exact	48	+8.3	[+3.1, +14.6]
GPT-5.4 mini (low)/canonical → state-canonical	future joint	48	+52.1	[+42.7, +61.5]
GPT-5.4 mini (high)/raw → state-raw	stock exact	48	+4.2	[-8.3, +16.7]
GPT-5.4 mini (high)/raw → state-raw	future joint	48	+13.5	[+0.0, +27.1]
GPT-5.4 mini (high)/canonical → state-canonical	stock exact	48	-1.0	[-11.5, +9.4]
GPT-5.4 mini (high)/canonical → state-canonical	future joint	48	+5.2	[-7.3, +17.7]
GPT-5.4 (low)/raw → state-raw	stock exact	48	+0.0	[-8.3, +8.3]
GPT-5.4 (low)/raw → state-raw	future joint	48	+0.0	[-8.3, +8.3]
GPT-5.4 (low)/canonical → state-canonical	stock exact	48	-1.0	[-3.1, +0.0]
GPT-5.4 (low)/canonical → state-canonical	future joint	48	+7.3	[+2.1, +13.5]
Sonnet 4.6/raw → state-raw	stock exact	48	-1.0	[-3.1, +0.0]
Sonnet 4.6/raw → state-raw	future joint	48	+0.0	[-3.1, +3.1]
Sonnet 4.6/canonical → state-canonical	stock exact	48	+0.0	[+0.0, +0.0]
Sonnet 4.6/canonical → state-canonical	future joint	48	+1.0	[+0.0, +3.1]
Sonnet + thinking/raw → state-raw	stock exact	48	+1.0	[+0.0, +3.1]
Sonnet + thinking/raw → state-raw	future joint	48	+2.1	[+0.0, +6.2]
Sonnet + thinking/canonical → state-canonical	stock exact	48	+0.0	[+0.0, +0.0]
Sonnet + thinking/canonical → state-canonical	future joint	48	+0.0	[+0.0, +0.0]
GPT-5.4 mini (low)/raw → tool-raw	future joint	48	+77.1	[+68.8, +84.4]
GPT-5.4 mini (low)/raw → tool-raw	stock exact	48	+75.0	[+66.7, +83.3]
GPT-5.4 mini (low)/state-raw → tool-raw	future joint	48	+71.9	[+60.4, +82.3]
GPT-5.4 mini (low)/canonical → tool-canonical	future joint	48	+53.1	[+43.8, +61.5]
GPT-5.4 mini (low)/canonical → tool-canonical	stock exact	48	+9.4	[+4.2, +16.7]
GPT-5.4 mini (low)/state-canonical → tool-canonical	future joint	48	+1.0	[+0.0, +3.1]
GPT-5.4 (low)/raw → tool-raw	future joint	48	+9.4	[+4.2, +15.6]
GPT-5.4 (low)/raw → tool-raw	stock exact	48	+9.4	[+4.2, +15.6]
GPT-5.4 (low)/state-raw → tool-raw	future joint	48	+9.4	[+4.2, +15.6]
GPT-5.4 (low)/canonical → tool-canonical	future joint	48	+8.3	[+3.1, +13.5]
GPT-5.4 (low)/canonical → tool-canonical	stock exact	48	+0.0	[+0.0, +0.0]
GPT-5.4 (low)/state-canonical → tool-canonical	future joint	48	+1.0	[+0.0, +3.1]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (low)/tool-raw	future joint	48	+75.0	[+64.6, +84.4]
GPT-5.4 mini (low, auto tool choice)/raw → GPT-5.4 mini (low)/tool-raw	stock exact	48	+70.8	[+60.4, +81.2]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (low)/tool-canonical	future joint	48	+17.7	[+10.4, +25.0]
GPT-5.4 mini (low, auto tool choice)/canonical → GPT-5.4 mini (low)/tool-canonical	stock exact	48	+7.3	[+3.1, +12.5]

Table 11: Paired differences in percentage points, retail profiles cohort: matched by history and repetition index, averaged within history; descriptive 95% history-bootstrap percentile intervals (10,000 draws), not multiplicity adjusted. The same-period replication moved one identical cell by -8.9 points, so differences of that size are within run-to-run variation. Cells at or near 100% give degenerate intervals. Identical rows arise when two comparisons have identical per-history differences.

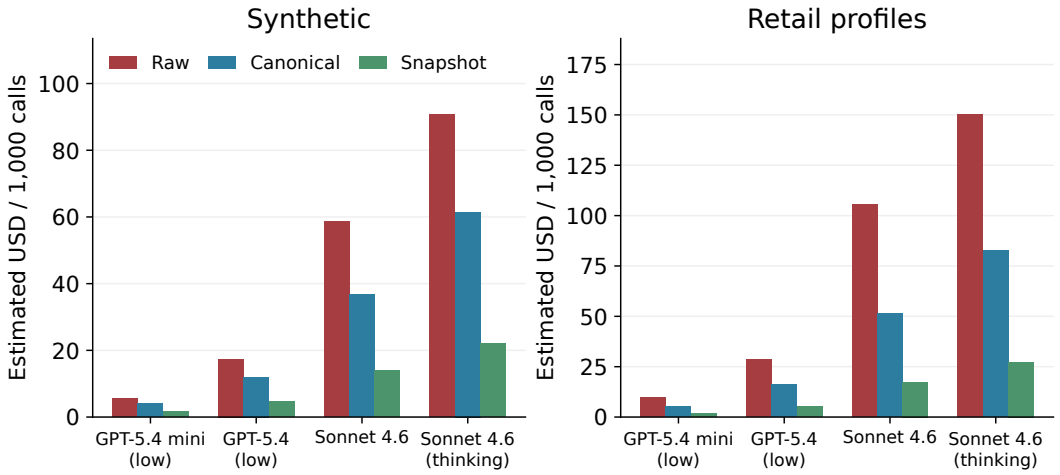


Figure 3: Observed API token-cost estimates under each representation in the original grid. These are configuration-dependent resource measurements, not compute-matched comparisons or provider billing invoices.

F Amendment 4: capability ladder, interfaces and transport

Protocol `protocol_v4_extensions.md` (frozen before its calls, with transport notes appended before each rerun) adds interface-matched GPT-5.4 cells (E6), Claude Haiku 4.5 (E7) and six open-weight models in a requested-effort by tool-choice cross (E8), plus a same-period GPT-5.4 forced-call replication (E6b). Open-weight models are served through OpenRouter with one pinned upstream per model and fallbacks disabled; the served provider is stored in every record and checked by the audit. Rate limiting by third-party upstreams (HTTP 429) is retried with bounded backoff before any completion exists, and requests that exhaust their attempts are re-issued in documented completion passes; a model output is never re-sampled. Two grids were interrupted and repeated in full before any accuracy was inspected, and the Qwen provider withdrew the model before the forced-call profile stage, so those four cells are absent.

Transport. Of the 23,808 Amendment 4 calls, 637 needed a bounded transport retry (HTTP 429/5xx before any completion existed; attempt counts are stored per record and no model output was re-sampled), and 0 open-weight calls were served by a provider other than the pinned one. Requested effort for the open-weight models is OpenRouter's unified `reasoning.effort`, which each upstream maps onto its own control.

Same-period GPT-5.4 forced-call replication (E6b). Future-joint accuracy on synthetic histories, snapshot: original 99.0, same-period replication 97.9, automatic choice 91.1%; state-canonical: original 98.4, same-period replication 99.0, automatic choice 83.9%. Paired drift, replication minus original forced cell: snapshot -1.0 [-3.6, +1.6]; state-canonical +0.5 [-1.6, +2.6]. Interface effect against the contemporaneous baseline, automatic minus replicated forced call: snapshot -6.8 [-11.5, -2.6]; state-canonical -15.1 [-21.4, -9.4].

The original forced-call placement table and the mini/GPT-5.4 requested-effort table follow for comparison with the interface-matched Table 1.

Model	Configuration	Arm	Future joint	Missing	Truncated	Mean reasoning tok.	Retried calls
GPT-5.4 mini	low, forced	raw	37.5	0	0	790	
	low, forced	canonical	76.0	0	0	515	
	high, forced	raw	25.0	0	0	1214	
	high, forced	canonical	34.9	0	0	708	
	low, auto	raw	56.8	1	0	975	
	low, auto	canonical	86.5	0	0	565	
	high, auto	raw	79.2	1	1	3054	
	high, auto	canonical	93.2	0	0	1425	
GPT-5.4	low, forced	raw	88.0	0	0	710	
	low, forced	canonical	98.4	0	0	419	
	high, forced	raw	94.3	0	0	1602	
	high, forced	canonical	94.8	0	0	749	
	low, auto	raw	98.4	0	0	707	
	low, auto	canonical	98.4	0	0	415	
	high, auto	raw	99.0	0	0	1554	
	high, auto	canonical	100.0	0	0	688	
gpt-oss-20b	low, forced	raw	23.4	85	0	494	
	low, forced	canonical	52.1	59	0	482	
	high, forced	raw	42.2	106	0	10545	
	high, forced	canonical	66.1	65	0	9862	
	low, auto	raw	24.0	88	0	498	
	low, auto	canonical	51.0	51	0	598	
	high, auto	raw	39.1	112	97	12678	
	high, auto	canonical	59.4	75	58	11037	
gpt-oss-120b	low, forced	raw	45.8	0	0	941	
	low, forced	canonical	69.3	0	0	664	
	high, forced	raw	53.6	89	89	6539	
	high, forced	canonical	66.1	65	65	5925	
	low, auto	raw	49.5	41	0	1023	
	low, auto	canonical	73.4	12	0	713	
	high, auto	raw	49.0	98	98	6679	
	high, auto	canonical	68.8	60	60	5986	
Qwen3.8-27B	low, forced	raw	76.6	40	0	2379	
	low, forced	canonical	91.7	16	0	1607	
	high, forced	raw	80.2	28	0	2103	
	high, forced	canonical	91.7	10	0	1225	
	low, auto	raw	95.3	3	1	2679	
	low, auto	canonical	99.5	0	0	1683	
	high, auto	raw	88.5	19	4	2271	
	high, auto	canonical	92.7	8	0	1212	
DeepSeek V4 Flash	low, forced	raw	2.1	4	0	103	
	low, forced	canonical	2.1	6	0	32	
	high, forced	raw	1.6	10	0	78	
	high, forced	canonical	1.0	15	0	24	
	low, auto	raw	98.4	0	0	3443	
	low, auto	canonical	99.0	0	0	1695	
	high, auto	raw	93.2	2	2	4756	
	high, auto	canonical	97.4	0	0	2451	
GLM-5.3 Flash	low, forced	raw	52.6	0	0	444	
	low, forced	canonical	62.5	0	0	316	
	high, forced	raw	74.5	0	0	823	
	high, forced	canonical	82.3	0	0	415	
	low, auto	raw	55.7	0	0	441	
	low, auto	canonical	64.1	2	1	441	
	high, auto	raw	76.6	1	0	991	
	high, auto	canonical	84.4	0	0	503	
DeepSeek V4 Pro	low, forced	raw	96.4	6	0	3662	
	low, forced	canonical	98.4	2	0	2523	
	high, forced	raw	91.7	16	9	5211	
	high, forced	canonical	82.3	32	1	3191	
	low, auto	raw	99.0	0	0	4142	
	low, auto	canonical	99.0	0	0	2461	
	high, auto	raw	93.2	12	11	5768	
	high, auto	canonical	97.4	2	2	3971	

Table 12: Effort $\times$ interface cells on the synthetic cohort (192 calls each): future-joint accuracy (%), calls with no answer (including refusals), calls truncated at the 16,384-token output limit, and mean provider-reported reasoning tokens. Each upstream maps the unified effort parameter onto its own control, so effort levels are comparable within a model family only.

Cohort	Configuration	Arm	refusal	text only	other tool	multiple calls	stream error	truncated	empty
Synthetic	DeepSeek V4 Flash (high, forced call)	canonical	0	0	0	11	0	4	0
Synthetic	DeepSeek V4 Flash (high, forced call)	raw	0	0	0	8	0	2	0
Synthetic	DeepSeek V4 Flash (low, forced call)	canonical	0	0	0	6	0	0	0
Synthetic	DeepSeek V4 Flash (low, forced call)	raw	0	0	0	4	0	0	0
Synthetic	DeepSeek V4 Pro (high, forced call)	canonical	0	0	0	0	0	1	31
Synthetic	DeepSeek V4 Pro (high, forced call)	raw	0	0	0	0	0	9	7
Synthetic	DeepSeek V4 Pro (high, auto tool choice)	raw	0	0	0	0	0	12	0
Synthetic	DeepSeek V4 Pro (low, forced call)	raw	0	0	0	0	0	0	6
Synthetic	Haiku 4.5 + thinking	raw	0	0	0	0	0	10	0
Synthetic	Haiku 4.5 + thinking	state-raw	0	0	0	0	0	11	0
Synthetic	gpt-oss-120b (high, forced call)	canonical	0	0	0	0	0	65	0
Synthetic	gpt-oss-120b (high, forced call)	raw	0	0	0	0	0	89	0
Synthetic	gpt-oss-120b (high, auto tool choice)	canonical	0	0	0	0	0	60	0
Synthetic	gpt-oss-120b (high, auto tool choice)	raw	0	0	0	0	0	98	0
Synthetic	gpt-oss-120b (low, auto tool choice)	canonical	7	5	0	0	0	0	0
Synthetic	gpt-oss-120b (low, auto tool choice)	raw	24	17	0	0	0	0	0
Synthetic	gpt-oss-120b (low, auto tool choice)	state-canonical	1	3	0	0	0	0	0
Synthetic	gpt-oss-120b (low, auto tool choice)	state-raw	9	7	0	0	0	0	0
Synthetic	gpt-oss-20b (high, forced call)	canonical	0	0	9	0	56	0	0
Synthetic	gpt-oss-20b (high, forced call)	raw	0	0	10	0	96	0	0
Synthetic	gpt-oss-20b (high, auto tool choice)	canonical	0	2	12	0	2	58	1
Synthetic	gpt-oss-20b (high, auto tool choice)	raw	0	1	11	0	0	97	3
Synthetic	gpt-oss-20b (low, forced call)	canonical	0	0	7	0	52	0	0
Synthetic	gpt-oss-20b (low, forced call)	raw	0	0	7	0	78	0	0
Synthetic	gpt-oss-20b (low, auto tool choice)	canonical	37	1	5	0	8	0	0
Synthetic	gpt-oss-20b (low, auto tool choice)	raw	70	1	6	0	11	0	0
Synthetic	gpt-oss-20b (low, auto tool choice)	state-canonical	18	1	12	0	1	0	0
Synthetic	gpt-oss-20b (low, auto tool choice)	state-raw	27	2	19	0	5	0	0
Synthetic	Qwen3.8-27B (high, forced call)	canonical	0	0	0	0	10	0	0
Synthetic	Qwen3.8-27B (high, forced call)	raw	0	0	0	0	28	0	0
Synthetic	Qwen3.8-27B (high, auto tool choice)	canonical	0	0	0	0	8	0	0
Synthetic	Qwen3.8-27B (high, auto tool choice)	raw	0	0	0	0	15	4	0
Synthetic	Qwen3.8-27B (low, forced call)	canonical	0	0	0	0	16	0	0
Synthetic	Qwen3.8-27B (low, forced call)	raw	0	0	0	0	40	0	0
Synthetic	Qwen3.8-27B (low, auto tool choice)	raw	0	0	0	0	1	1	1
Synthetic	Qwen3.8-27B (low, auto tool choice)	state-raw	0	0	0	0	0	3	0
Retail profiles	DeepSeek V4 Flash (high, forced call)	raw	0	0	0	6	0	2	0
Retail profiles	DeepSeek V4 Flash (high, auto tool choice)	raw	0	0	2	0	0	1	0
Retail profiles	DeepSeek V4 Flash (low, forced call)	canonical	0	0	0	3	0	1	0
Retail profiles	DeepSeek V4 Pro (high, forced call)	canonical	0	0	0	0	0	1	19
Retail profiles	DeepSeek V4 Pro (high, forced call)	raw	0	0	0	0	0	4	15
Retail profiles	DeepSeek V4 Pro (high, auto tool choice)	raw	0	0	0	0	0	3	0
Retail profiles	DeepSeek V4 Pro (low, forced call)	raw	0	0	0	0	0	0	7
Retail profiles	Haiku 4.5 + thinking	raw	0	0	0	0	0	37	0
Retail profiles	Haiku 4.5 + thinking	state-raw	0	0	0	0	0	21	0
Retail profiles	gpt-oss-120b (high, forced call)	canonical	0	0	1	0	0	84	0
Retail profiles	gpt-oss-120b (high, forced call)	raw	0	0	0	0	0	96	0
Retail profiles	gpt-oss-120b (high, auto tool choice)	canonical	0	0	0	0	0	86	0
Retail profiles	gpt-oss-120b (high, auto tool choice)	raw	0	0	0	0	0	96	0
Retail profiles	gpt-oss-120b (low, auto tool choice)	raw	17	30	0	0	0	0	0
Retail profiles	gpt-oss-20b (high, forced call)	canonical	0	0	1	0	48	0	0
Retail profiles	gpt-oss-20b (high, forced call)	raw	0	0	0	0	96	0	0
Retail profiles	gpt-oss-20b (high, auto tool choice)	canonical	0	0	3	0	1	46	0
Retail profiles	gpt-oss-20b (high, auto tool choice)	raw	0	0	0	0	0	96	0
Retail profiles	gpt-oss-20b (low, forced call)	canonical	0	0	9	0	44	0	0
Retail profiles	gpt-oss-20b (low, forced call)	raw	0	0	6	0	67	0	0
Retail profiles	gpt-oss-20b (low, auto tool choice)	canonical	44	2	7	0	5	0	0
Retail profiles	gpt-oss-20b (low, auto tool choice)	raw	55	2	8	0	4	0	0

Table 13: Missing answers by kind for every cell with at least three (all counted as failures in every accuracy): *refusal*, a text reply declining the task; *text only*, an answer written in the message instead of the tool call; *other tool* and *multiple calls*, calls to a non-existent tool or more than one call; *stream error*, a provider-side error after the request was accepted; *truncated*, the 16,384-token output limit reached (for Anthropic thinking runs, entirely in thinking); *empty*, no content.

Cohort	Model, effort	Arm	Automatic − forced (future joint)	Automatic − forced (stock exact)
Synthetic	GPT-5.4 mini, low	raw	+19.3 [+11.5, +27.1]	+8.3 [+1.6, +15.1]
Synthetic	GPT-5.4 mini, high	raw	+54.2 [+44.3, +63.5]	+40.6 [+31.8, +49.5]
Synthetic	GPT-5.4 mini, low	canonical	+10.4 [+3.1, +17.7]	+3.6 [-1.6, +8.9]
Synthetic	GPT-5.4 mini, high	canonical	+58.3 [+49.5, +66.7]	+40.6 [+31.8, +49.5]
Synthetic	GPT-5.4, low	raw	+10.4 [+5.2, +16.1]	+7.3 [+2.6, +12.5]
Synthetic	GPT-5.4, high	raw	+4.7 [+1.0, +8.3]	+2.6 [+0.0, +5.2]
Synthetic	GPT-5.4, low	canonical	+0.0 [-2.6, +2.6]	-1.0 [-3.1, +1.0]
Synthetic	GPT-5.4, high	canonical	+5.2 [+2.1, +8.9]	+1.6 [+0.0, +3.6]
Synthetic	gpt-oss-20b, low	raw	+0.5 [-5.7, +6.8]	+0.5 [-6.2, +7.3]
Synthetic	gpt-oss-20b, high	raw	-3.1 [-8.9, +2.1]	-1.6 [-6.8, +3.6]
Synthetic	gpt-oss-20b, low	canonical	-1.0 [-8.3, +6.2]	+4.2 [-2.1, +10.4]
Synthetic	gpt-oss-20b, high	canonical	-6.8 [-13.5, +0.0]	-5.7 [-12.5, +1.0]
Synthetic	gpt-oss-120b, low	raw	+3.6 [-3.1, +10.4]	+1.0 [-4.7, +6.8]
Synthetic	gpt-oss-120b, high	raw	-4.7 [-8.9, -1.0]	-4.7 [-8.9, -1.0]
Synthetic	gpt-oss-120b, low	canonical	+4.2 [-3.1, +12.0]	-1.6 [-7.3, +4.2]
Synthetic	gpt-oss-120b, high	canonical	+2.6 [-2.6, +7.8]	+2.6 [-2.6, +7.8]
Synthetic	Qwen3.8-27B, low	raw	+18.8 [+12.5, +25.5]	+18.2 [+12.0, +25.0]
Synthetic	Qwen3.8-27B, high	raw	+8.3 [+1.6, +15.1]	+6.8 [+0.0, +13.5]
Synthetic	Qwen3.8-27B, low	canonical	+7.8 [+4.2, +12.0]	+7.8 [+4.2, +12.0]
Synthetic	Qwen3.8-27B, high	canonical	+1.0 [-4.7, +6.8]	+1.6 [-3.1, +5.7]
Synthetic	DeepSeek V4 Flash, low	raw	+96.4 [+93.8, +99.0]	+92.2 [+88.0, +95.8]
Synthetic	DeepSeek V4 Flash, high	raw	+91.7 [+87.0, +95.8]	+87.5 [+81.8, +92.7]
Synthetic	DeepSeek V4 Flash, low	canonical	+96.9 [+94.3, +99.0]	+91.1 [+87.0, +94.8]
Synthetic	DeepSeek V4 Flash, high	canonical	+96.4 [+93.2, +99.0]	+94.3 [+90.1, +97.9]
Synthetic	GLM-5.3 Flash, low	raw	+3.1 [-6.2, +12.5]	-2.1 [-9.4, +4.7]
Synthetic	GLM-5.3 Flash, high	raw	+2.1 [-5.2, +9.4]	+1.0 [-3.6, +5.7]
Synthetic	GLM-5.3 Flash, low	canonical	+1.6 [-6.8, +9.9]	+1.6 [-5.2, +8.3]
Synthetic	GLM-5.3 Flash, high	canonical	+2.1 [-4.7, +8.9]	+2.1 [-2.1, +6.2]
Synthetic	DeepSeek V4 Pro, low	raw	+2.6 [-0.5, +5.7]	+3.1 [+0.0, +6.2]
Synthetic	DeepSeek V4 Pro, high	raw	+1.6 [-2.6, +5.7]	+2.1 [-1.6, +5.7]
Synthetic	DeepSeek V4 Pro, low	canonical	+0.5 [-1.6, +2.6]	+1.0 [+0.0, +2.6]
Synthetic	DeepSeek V4 Pro, high	canonical	+15.1 [+9.4, +20.8]	+14.1 [+8.3, +19.8]
Retail profiles	GPT-5.4 mini, low	raw	+2.1 [-6.2, +11.5]	+4.2 [-4.2, +13.5]
Retail profiles	GPT-5.4 mini, high	raw	+28.1 [+15.6, +39.6]	+18.8 [+7.3, +30.2]
Retail profiles	GPT-5.4 mini, low	canonical	+35.4 [+22.9, +47.9]	+2.1 [-6.2, +10.4]
Retail profiles	GPT-5.4 mini, high	canonical	+65.6 [+55.2, +75.0]	+61.5 [+50.0, +71.9]
Retail profiles	GPT-5.4, low	raw	+7.3 [+3.1, +12.5]	+7.3 [+3.1, +12.5]
Retail profiles	GPT-5.4, high	raw	+1.0 [+0.0, +3.1]	+1.0 [+0.0, +3.1]
Retail profiles	GPT-5.4, low	canonical	+8.3 [+3.1, +13.5]	+0.0 [+0.0, +0.0]
Retail profiles	GPT-5.4, high	canonical	+11.5 [+6.2, +17.7]	+9.4 [+4.2, +15.6]
Retail profiles	gpt-oss-20b, low	raw	-1.0 [-3.1, +0.0]	-1.0 [-7.3, +4.2]
Retail profiles	gpt-oss-20b, high	raw	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Retail profiles	gpt-oss-20b, low	canonical	+6.2 [-3.1, +15.6]	+2.1 [-8.3, +12.5]
Retail profiles	gpt-oss-20b, high	canonical	-1.0 [-14.6, +12.5]	+0.0 [-13.5, +13.5]
Retail profiles	gpt-oss-120b, low	raw	-2.1 [-8.3, +3.1]	-7.3 [-13.6, -1.0]
Retail profiles	gpt-oss-120b, high	raw	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Retail profiles	gpt-oss-120b, low	canonical	-3.1 [-14.6, +9.4]	-2.1 [-11.5, +8.3]
Retail profiles	gpt-oss-120b, high	canonical	-1.0 [-10.4, +8.3]	-1.0 [-10.4, +8.3]
Retail profiles	DeepSeek V4 Flash, low	raw	+94.8 [+88.5, +99.0]	+87.5 [+79.2, +94.8]
Retail profiles	DeepSeek V4 Flash, high	raw	+93.8 [+87.5, +99.0]	+89.6 [+80.2, +96.9]
Retail profiles	DeepSeek V4 Flash, low	canonical	+95.8 [+91.7, +99.0]	+87.5 [+79.2, +94.8]
Retail profiles	DeepSeek V4 Flash, high	canonical	+97.9 [+94.8, +100.0]	+88.5 [+80.2, +95.8]
Retail profiles	GLM-5.3 Flash, low	raw	-5.2 [-18.8, +8.3]	-2.1 [-15.6, +11.5]
Retail profiles	GLM-5.3 Flash, high	raw	+0.0 [-11.5, +11.5]	-3.1 [-11.5, +5.2]
Retail profiles	GLM-5.3 Flash, low	canonical	-6.2 [-20.8, +7.3]	-3.1 [-12.5, +5.2]
Retail profiles	GLM-5.3 Flash, high	canonical	+1.0 [-8.3, +10.4]	+0.0 [-6.2, +6.2]
Retail profiles	DeepSeek V4 Pro, low	raw	+4.2 [-3.1, +11.5]	+6.2 [+0.0, +12.5]
Retail profiles	DeepSeek V4 Pro, high	raw	+17.7 [+9.4, +26.0]	+17.7 [+9.4, +26.0]
Retail profiles	DeepSeek V4 Pro, low	canonical	+3.1 [+0.0, +7.3]	+2.1 [+0.0, +5.2]
Retail profiles	DeepSeek V4 Pro, high	canonical	+21.9 [+14.6, +29.2]	+21.9 [+14.6, +29.2]

Table 14: Interface effect: automatic tool choice minus forced answer call at fixed model, effort, input and history; paired differences in percentage points with descriptive 95% history-bootstrap intervals.

	Forced answer call			Automatic tool choice			
Model	low	high	high−low	low	high	high−low	Interaction
GPT-5.4 mini	37.5	25.0	-12.5	56.8	79.2	+22.4	+34.9 [+22.4, +46.9]
GPT-5.4	88.0	94.3	+6.2	98.4	99.0	+0.5	-5.7 [-12.5, +1.0]
gpt-oss-20b	23.4	42.2	+18.8	24.0	39.1	+15.1	-3.6 [-12.0, +4.7]
gpt-oss-120b	45.8	53.6	+7.8	49.5	49.0	-0.5	-8.3 [-16.7, -0.5]
Qwen3.8-27B	76.6	80.2	+3.6	95.3	88.5	-6.8	-10.4 [-20.3, -0.5]
DeepSeek V4 Flash	2.1	1.6	-0.5	98.4	93.2	-5.2	-4.7 [-9.9, +0.5]
GLM-5.3 Flash	52.6	74.5	+21.9	55.7	76.6	+20.8	-1.0 [-13.0, +10.9]
DeepSeek V4 Pro	96.4	91.7	-4.7	99.0	93.2	-5.7	-1.0 [-6.2, +3.6]

Table 15: Requested reasoning effort × tool-choice interface, synthetic cohort, raw input, future-joint accuracy (%; 192 calls per cell). high−low: paired effort effect within an interface. Interaction: automatic-choice effort effect minus forced-call effort effect, paired by history, with a descriptive 95% history-bootstrap interval (10,000 draws); a positive interaction means the forced call turns extra effort into a liability. Requested effort is a family-specific provider setting, not a compute match (observed reasoning tokens: Appendix F).

	Raw input			Canonical input			Snapshot
Cohort, configuration	model	hybrid	inside	model	hybrid	inside	model
Synthetic, GPT-5.4 mini (low)	37.5	55.2	91.7	76.0	93.8	97.9	89.6
Synthetic, GPT-5.4 (low)	88.0	91.7	100.0	98.4	98.4	97.4	99.0
Synthetic, Sonnet 4.6	99.0	100.0	–	99.5	100.0	–	100.0
Synthetic, Sonnet + thinking	97.9	100.0	–	100.0	100.0	–	100.0
Retail profiles, GPT-5.4 mini (low)	12.5	17.7	89.6	46.9	99.0	100.0	89.6
Retail profiles, GPT-5.4 (low)	90.6	90.6	100.0	91.7	99.0	100.0	99.0
Retail profiles, Sonnet 4.6	95.8	95.8	–	99.0	100.0	–	100.0
Retail profiles, Sonnet + thinking	97.9	100.0	–	100.0	100.0	–	100.0

Table 16: Future-joint accuracy (%, all three scenario outputs exact) by input and code placement; 192 synthetic and 96 profile calls per cell; invalid answers fail. *model*: all five fields from the model; *hybrid*: model returns the two historical fields, code applies the schedule (equals reconstruction accuracy by conservation); *inside*: code-execution tool available during the call (OpenAI only); *snapshot*: exact state supplied, model rolls out; dashes mark arms not run. All-five accuracy: Appendix B.

		Synthetic			Retail profiles		
Model	Arm	low	medium	high	low	medium	high
GPT-5.4 mini, forced call	Raw	37.5	26.6	25.0	12.5	16.7	20.8
	Canonical	76.0	46.9	34.9	46.9	38.5	29.2
	Snapshot	89.6	97.9	97.4	89.6	94.8	93.8
GPT-5.4, forced call	Raw	88.0	95.3	94.3	90.6	95.8	99.0
	Canonical	98.4	95.8	94.8	91.7	94.8	88.5
GPT-5.4 mini, auto choice	Raw	56.8	–	79.2	14.6	–	49.0
	Canonical	86.5	–	93.2	82.3	–	94.8

Table 17: Future-joint accuracy (%) by requested reasoning effort, identical prompts and tasks; 192 synthetic and 96 profile calls per cell; the forced-call low columns are the original cohorts. Forced call: the answer function is the only permitted action; auto choice: the model may respond freely and a missing answer fails. Dashes: not run.